

Indian Institute of Technology Guwahati

Department of Electrical and Electronics Engineering
System Control and Automation

REPORT

**Control and Simulation of a Bidirectional
DC/DC Converter
for a 5-kW Distributed Wind-Solar Hybrid
Energy System**

A Simulation-Based Study of Bidirectional Power Flow, Converter Control, Battery SOC
Estimation and Energy Management

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Chapter 1

Introduction

1.1 Background

The increasing penetration of renewable energy sources has created a strong need for efficient and reliable power-conditioning systems. Among various renewable-energy technologies, solar photovoltaic (PV) and wind energy are attractive because they can be used independently or in combination to supply electrical power. However, both sources are inherently variable. Solar power changes with solar irradiance and time of day, while wind power varies with wind speed. Consequently, the instantaneous power generated by a renewable-energy system may not always match the power demanded by the connected load.

A **wind-solar hybrid energy system** combines the complementary characteristics of wind and solar generation to improve the availability of renewable power. Such systems are particularly useful in remote or distributed applications where connection to a conventional utility grid may be difficult or uneconomical. The reference work specifically considers a small-capacity distributed wind-solar hybrid system intended for remote applications, where the converter is required to have relatively small size, low weight and convenient portability.

The basic challenge in such a system is the mismatch between **generated power** and **load power**. Let

$$P_G \tag{1.1}$$

denote the instantaneous power generated by the renewable sources and

$$P_L \tag{1.2}$$

denote the instantaneous load demand.

Depending on operating conditions, three situations can occur.

When

$$P_G > P_L, \tag{1.3}$$

the renewable sources are generating more power than required by the load. The excess energy can therefore be stored in the battery.

When

$$P_G = P_L, \tag{1.4}$$

the generated power is approximately equal to the load demand, and the battery ideally does not need to exchange significant power with the DC bus.

Finally, when

$$P_G < P_L, \tag{1.5}$$

a power deficit occurs. In this condition, the battery must supply the required additional power to maintain the energy balance of the system.

Therefore, an energy-storage system is required to compensate for the difference between renewable generation and load demand. Battery storage is particularly important for handling sudden changes in generation and load and for maintaining smoother system operation.

1.2 Wind-Solar Hybrid Energy System

A typical wind-solar hybrid system consists of a **wind energy conversion unit**, a **solar PV generation unit**, a common **DC bus**, an electrical load, and an **energy-storage system**.

The renewable sources are connected to the DC bus through appropriate power-electronic conversion stages. The DC bus acts as the common electrical interface through which power is transferred between the renewable sources, load and energy-storage system.

1.2.1 Overall Energy Flow

The overall energy flow of the system is illustrated in Fig. 1.1.

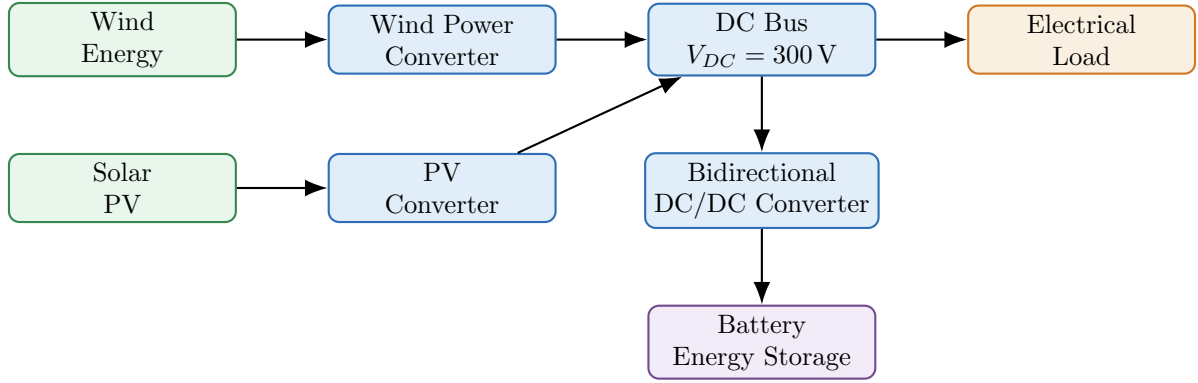


Figure 1.1: Conceptual architecture of the wind-solar hybrid energy-storage system.

The bidirectional DC/DC converter is placed between the **DC bus and battery**, allowing electrical energy to flow in either direction.

This architecture provides two important operating modes.

1.2.2 Charging Mode

When

$$P_G > P_L, \quad (1.6)$$

the renewable sources generate more power than the load requires. The excess renewable power is therefore transferred from the DC bus to the battery.

$$\boxed{P_G > P_L \Rightarrow \text{Battery Charging}} \quad (1.7)$$

The corresponding power-flow direction is

$$\text{Renewable Sources} \rightarrow \text{DC Bus} \rightarrow \text{Bidirectional DC/DC Converter} \rightarrow \text{Battery}. \quad (1.8)$$

1.2.3 Discharging Mode

When

$$P_G < P_L, \quad (1.9)$$

the battery supplies additional power through the bidirectional converter to the DC bus.

$$\boxed{P_G < P_L \Rightarrow \text{Battery Discharging}} \quad (1.10)$$

The corresponding power-flow direction is

$$\text{Battery} \rightarrow \text{Bidirectional DC/DC Converter} \rightarrow \text{DC Bus} \rightarrow \text{Load}. \quad (1.11)$$

Thus, the bidirectional converter becomes the key interface responsible for managing the exchange of energy between the battery and the DC bus.

1.3 Need for Energy Storage

The output of a renewable-energy system is not constant. During periods of high renewable generation, the generated power may exceed the load requirement. Without an energy-storage system, this excess power may not be effectively utilized.

Conversely, during periods of low wind or solar generation, the available renewable power may be insufficient to meet the load demand.

The battery therefore performs an **energy-balancing function**:

$$\boxed{P_G - P_L > 0 \Rightarrow \text{Battery Charging}} \quad (1.12)$$

and

$$\boxed{P_G - P_L < 0 \Rightarrow \text{Battery Discharging}} \quad (1.13)$$

However, the battery cannot simply be charged or discharged without limits. Excessive charging or deep discharge can adversely affect battery operation. Consequently, the battery **State of Charge (SOC)** must be monitored and incorporated into the energy-management strategy.

1.4 Role of the Bidirectional DC/DC Converter

A conventional unidirectional DC/DC converter can transfer energy in only one direction. Such a converter is insufficient for an energy-storage system because the battery must both absorb and supply energy.

A **bidirectional DC/DC converter** solves this problem by allowing power transfer in both directions between the DC bus and battery.

In the considered topology, the converter is a **non-isolated bidirectional DC/DC converter**. It consists of semiconductor switches, diodes, inductive and capacitive elements, and associated resistive components.

The converter performs two principal functions.

1.4.1 Step-Down Operation – Buck Mode

During buck operation, the DC-bus voltage is reduced to a suitable battery-side voltage.

$$\boxed{DC \text{ Bus} \rightarrow \text{Converter} \rightarrow \text{Battery}} \quad (1.14)$$

This mode is associated with **battery charging**.

1.4.2 Step-Up Operation – Boost Mode

During boost operation, the battery voltage is increased to the required DC-bus voltage.

$$\boxed{\text{Battery} \rightarrow \text{Converter} \rightarrow DC \text{ Bus}} \quad (1.15)$$

This mode is associated with **battery discharging**.

1.5 Operating Principle of the Converter

The two main semiconductor switches are denoted by

$$VT_1 \quad (1.16)$$

and

$$VT_2. \quad (1.17)$$

Their switching states determine the operating mode of the converter.

Table 1.1: Principal switching states of the bidirectional converter.

VT_1	VT_2	Operating Mode	Power Flow
OFF	ON	Buck / Step-down	DC bus $\rightarrow$ Battery
ON	OFF	Boost / Step-up	Battery $\rightarrow$ DC bus
OFF	OFF	Both switches OFF	No active switching path
ON	ON	Forbidden state	Not permitted

The simultaneous ON condition must be avoided because it can create an undesirable direct conduction path and cause improper converter operation. Therefore, switching logic is required to prevent VT_1 and VT_2 from turning ON simultaneously.

This switching arrangement is fundamental to the bidirectional nature of the converter.

1.6 DC-Bus Voltage Regulation

The DC bus is an important node in the hybrid system because it connects the renewable generation, load and energy-storage subsystem.

For the system considered in the reference work, the nominal DC-bus voltage is

$$\boxed{V_{DC}^* = 300 \text{ V}} \quad (1.18)$$

where V_{DC}^* denotes the desired or reference DC-bus voltage.

The stability of the DC-bus voltage is particularly important because variations in the bus voltage affect the energy flow between the generation system, load and battery.

The control objective can therefore be expressed as

$$\boxed{V_{DC}(t) \rightarrow V_{DC}^*} \quad (1.19)$$

The voltage error is defined as

$$e_v(t) = V_{DC}^* - V_{DC}(t). \quad (1.20)$$

The controller uses this error to modify the converter control signal and consequently regulate the power exchanged with the battery.

1.7 Control of the Bidirectional Converter

The bidirectional converter requires a suitable control strategy because its input and output voltages can vary during charging and discharging.

1.7.1 Buck-Mode Control

For buck operation, the ideal voltage-conversion relationship is

$$\boxed{D = \frac{U_o}{U_i}} \quad (1.21)$$

where

- D is the converter duty ratio,
- U_i is the converter input voltage, and
- U_o is the desired output voltage.

The voltage error is defined as

$$e(t) = U_o(t) - U'_o(t), \quad (1.22)$$

where $U'_o(t)$ represents the actual output voltage.

1.7.2 Boost-Mode Control

For boost operation, the ideal duty-ratio relationship is

$$D = 1 - \frac{U_i}{U_o} \quad (1.23)$$

where U_i is the converter input voltage and U_o is the desired output voltage. The desired and actual duty ratios may be expressed as

$$D^* = 1 - \frac{U_i}{U_o} \quad (1.24)$$

and

$$D' = 1 - \frac{U_i}{U'_o}. \quad (1.25)$$

Therefore, the duty-ratio error is

$$e(t) = D^*(t) - D'(t). \quad (1.26)$$

The control structure uses the calculated duty ratio to generate PWM switching signals.

The reference implementation uses a triangular carrier with an amplitude of 1 and switching frequency

$$f_{sw} = 20 \text{ kHz} \quad (1.27)$$

for PWM generation.

For boost operation, the calculated duty ratio is limited to

$$0 \leq D \leq 0.9. \quad (1.28)$$

This limitation prevents unsuitable operating conditions associated with the converter inductance and duty ratio.

1.8 Battery State-of-Charge Estimation

The **State of Charge (SOC)** represents the amount of charge available in the battery relative to its rated capacity.

In this work, the **Ampere-Hour method** is used for SOC estimation. The method determines the change in battery charge by integrating the battery charging and discharging current over time.

The general concept can be expressed as

$$SOC(t) = \frac{Q(t)}{Q_I} \quad (1.29)$$

where

- $SOC(t)$ is the battery State of Charge,
- $Q(t)$ is the available battery charge, and
- Q_I is the rated battery capacity.

The SOC information is then used by the energy-management algorithm to determine whether the battery can continue charging or discharging.

For the reference simulation, the battery capacity is

$$Q_I = 2 \text{ Ah} \quad (1.30)$$

with an initial charge of zero, charging efficiency of unity and zero self-discharge assumed for the idealized battery model.

1.9 Energy-Management Strategy

The energy-management system determines the operating state of the battery based on generated power, load power and SOC.

The fundamental decision process is illustrated in Fig. 2.5.

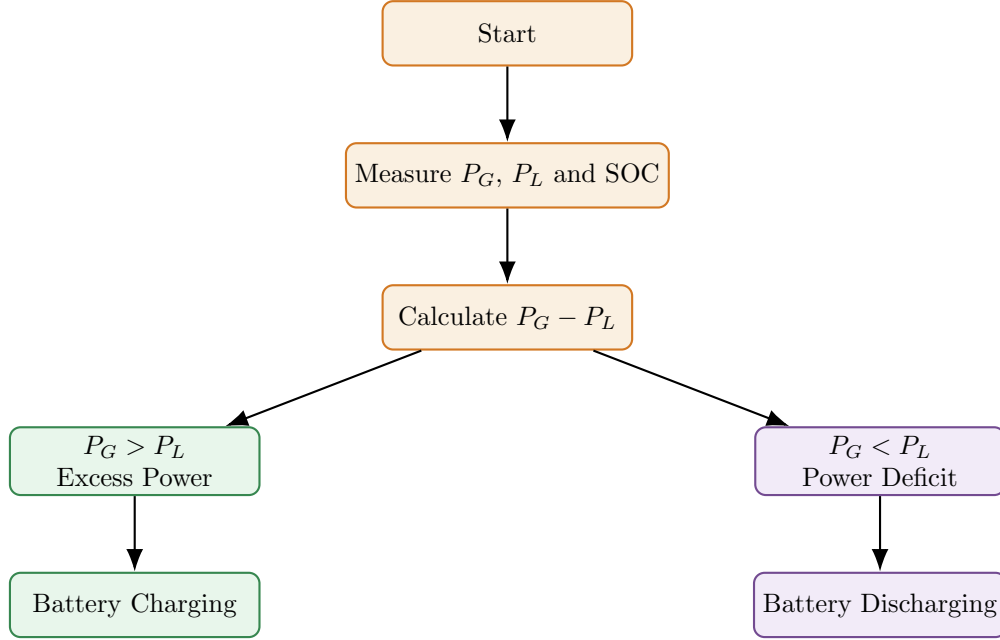


Figure 1.2: Simplified energy-management decision process.

The SOC-based control strategy constrains the charging and discharging operation using predefined SOC limits.

The principal thresholds used in the reference system are

$$SOC > 0.8 \quad (1.31)$$

and

$$SOC < 0.2. \quad (1.32)$$

These correspond approximately to 80% and 20% SOC limits, respectively.

1.10 MATLAB/Simulink-Based Simulation

The complete system is implemented and investigated using **MATLAB/Simulink**.

The simulation consists of the following principal subsystems:

Wind/Solar Generation → DC Bus → Bidirectional DC/DC → Battery → SOC Model → Energy Management
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(1.33)

The converter model includes both buck and boost operating modes, while the battery subsystem includes charging, discharging and SOC estimation.

The principal quantities investigated in the simulation are

$$V_B(t) \quad (1.34)$$

for battery voltage,

$$I_B(t) \quad (1.35)$$

for battery current,

$$SOC(t) \quad (1.36)$$

for battery State of Charge,

$$V_{DC}(t) \quad (1.37)$$

for DC-bus voltage, and

$$P_G(t), \quad P_L(t), \quad P_B(t) \quad (1.38)$$

for generated power, load power and battery power, respectively.

1.11 Motivation for the Present Study

Although the basic function of a bidirectional converter is relatively straightforward, its control becomes more challenging when the operating conditions change continuously.

The converter must simultaneously handle:

- variation in renewable generation,
- variation in load demand,
- changing battery voltage,
- charging and discharging transitions,
- SOC constraints,
- DC-bus voltage regulation,
- PWM generation, and
- safe switching of the semiconductor devices.

The bidirectional converter must therefore operate reliably under different combinations of generation, load and battery conditions.

This provides the motivation for studying the mathematical model, control strategy, battery SOC estimation and complete simulation of the bidirectional DC/DC converter within a 5-kW wind-solar hybrid energy system.

1.12 Problem Statement

Problem Statement

A wind-solar hybrid energy system experiences continuous variation in renewable generation and load demand, resulting in an imbalance between generated power and consumed power. An energy-storage system is therefore required to absorb excess renewable energy and supply power during generation deficits.

To achieve this bidirectional energy exchange, a suitable bidirectional DC/DC converter and associated control strategy are required. The converter must provide buck operation during battery charging and boost operation during battery discharging while maintaining the required DC-bus voltage and respecting battery SOC constraints.

The objective is therefore to develop and simulate a coordinated converter-control and energy-management system capable of achieving stable bidirectional power flow in a 5-kW wind-solar hybrid energy-storage system.

1.13 Objectives of the Work

The objectives of the report are divided into a primary objective and specific objectives.

1.13.1 Primary Objective

The primary objective is to model, control and simulate a **bidirectional DC/DC converter for a 5-kW wind-solar hybrid energy-storage system**.

1.13.2 Specific Objectives

The specific objectives are:

1. To study the architecture of a wind-solar hybrid energy system.
2. To develop the mathematical model of the bidirectional DC/DC converter.
3. To analyze the **buck operating mode** for battery charging.
4. To analyze the **boost operating mode** for battery discharging.
5. To develop the duty-ratio control equations for both operating modes.
6. To implement PWM generation at the selected switching frequency.
7. To develop appropriate switching logic for VT_1 and VT_2 .
8. To implement battery SOC estimation using the Ampere-Hour method.
9. To develop the battery charge/discharge energy-management algorithm.
10. To regulate the DC-bus voltage around its reference value.
11. To implement the complete system in MATLAB/Simulink.
12. To evaluate battery voltage, battery current, SOC, DC-bus voltage and power-flow responses.
13. To assess the effectiveness of the bidirectional converter in maintaining controlled energy exchange between the battery and DC bus.

1.14 Scope of the Report

The scope of the work is illustrated conceptually in Fig. 1.3.

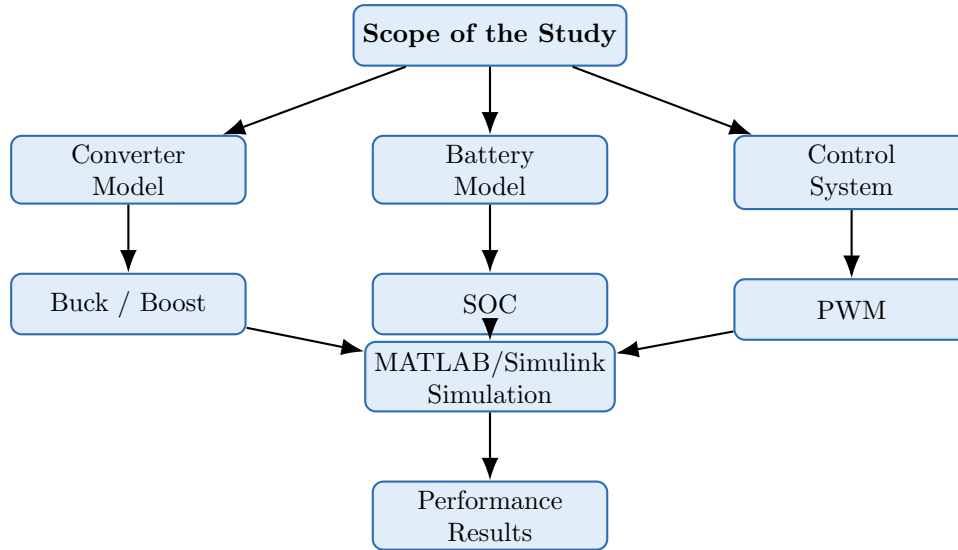


Figure 1.3: Scope of the present study.

The scope of the report includes the converter model, battery model, control system, SOC estimation, energy management, MATLAB/Simulink implementation and simulation analysis.

The primary focus is on the **simulation and control analysis** of the bidirectional converter and energy-storage subsystem.

If hardware implementation is not part of the present project, it should be clearly stated that experimental validation is outside the primary scope of this report.

Chapter 2

System Configuration

2.1 Overview of the System

The system considered in this work is a **5-kW distributed wind-solar hybrid energy system incorporating a battery energy-storage unit**. The main purpose of the system is to utilize renewable energy sources while maintaining a suitable power balance between renewable generation, energy storage and the electrical load.

The reference system is intended for distributed applications, particularly remote locations where a compact and lightweight energy-conversion system is desirable. To reduce the number of electronic components and circuit boards, the system employs a **bidirectional DC/DC converter** as the interface between the DC bus and the battery. :contentReference[oaicite:1]index=1

The complete system can be divided into the following major subsystems:

1. Wind-energy generation subsystem.
2. Solar-energy generation subsystem.
3. DC-bus subsystem.
4. Electrical load.
5. Battery energy-storage subsystem.
6. Bidirectional DC/DC converter.
7. Converter control and PWM subsystem.
8. Battery SOC monitoring and energy-management subsystem.

The interaction between these subsystems determines the direction and magnitude of power transferred between the renewable sources, load and battery.

2.2 Overall Wind-Solar Hybrid Architecture

The overall wind-solar hybrid system consists of two renewable energy sources, namely wind and solar photovoltaic generation, connected to a common DC bus. The DC bus supplies the electrical load and is connected to the battery through a bidirectional DC/DC converter.

The overall system architecture is shown in Fig. 2.1.

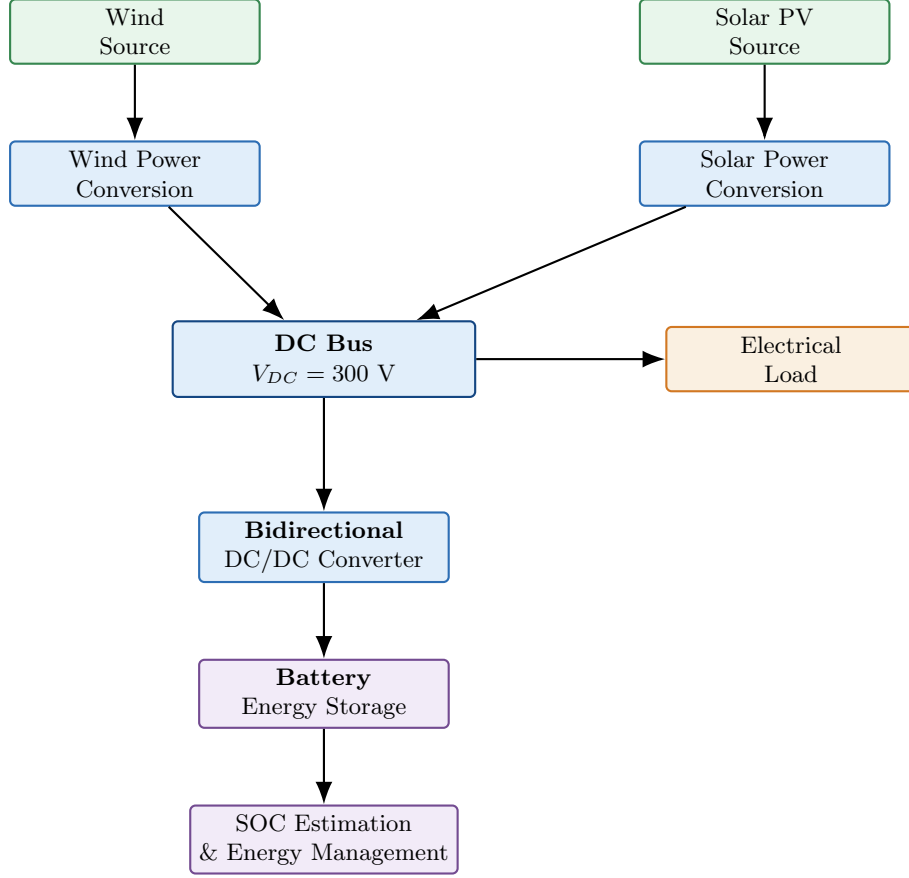


Figure 2.1: Overall configuration of the 5-kW wind-solar hybrid energy-storage system.

The source paper establishes the hybrid-system structure and places the bidirectional converter between the DC bus and the battery. The DC bus therefore acts as the central electrical interface for power exchange between renewable generation, the electrical load and the battery. :contentReference[oaicite:2]index=2

2.2.1 Main System Notations

The principal power and voltage quantities used throughout this chapter are defined as

$$P_G = \text{Total renewable generated power} \quad (2.1)$$

$$P_L = \text{Load power} \quad (2.2)$$

$$P_B = \text{Battery power} \quad (2.3)$$

and

$$V_{DC} = \text{DC-bus voltage.} \quad (2.4)$$

The generated power can further be expressed as

$$P_G = P_{\text{wind}} + P_{\text{solar}} \quad (2.5)$$

where P_{wind} represents the wind-generated power and P_{solar} represents the solar-generated power. :contentReference[oaicite:3]index=3

2.3 Renewable Energy Generation Subsystem

The renewable-generation subsystem consists of the **wind-energy source** and the **solar photovoltaic source**. Their purpose is to provide renewable electrical power to the common DC bus.

Since renewable generation changes with operating conditions, the instantaneous generated power is not necessarily equal to the load demand.

The total renewable generation is represented by

$$P_G = P_{\text{wind}} + P_{\text{solar}} \quad (2.6)$$

where

- P_{wind} is the power generated by the wind-energy subsystem, and
- P_{solar} is the power generated by the solar PV subsystem.

This equation provides a system-level representation of renewable generation. The detailed wind-turbine and PV mathematical models are outside the scope of the source document and should be introduced separately only if they are developed as part of the present work. The uploaded material explicitly makes this distinction. :contentReference[oaicite:4]index=4

2.4 DC Bus

The DC bus acts as the **central electrical node** of the hybrid energy system. It connects the renewable-generation units, the electrical load and the battery energy-storage system through the bidirectional DC/DC converter.

The nominal DC-bus voltage in the reference system is

$$V_{DC} = 300 \text{ V} \quad (2.7)$$

The DC bus is particularly important because it provides a common voltage level for power exchange between the different subsystems. The reference paper specifies a 300-V DC bus and emphasizes its importance in charging, discharging and overall energy-flow operation. :contentReference[oaicite:5]index=5

2.4.1 DC-Bus Power Balance

At the system level, the instantaneous power balance can be examined using

$$P_G - P_L. \quad (2.8)$$

Three principal operating conditions are possible.

Case 1: Excess Renewable Generation

When

$$P_G > P_L \quad (2.9)$$

the renewable generation is greater than the load demand.

Consequently, excess energy is available for storage:

$$P_G - P_L > 0 \quad \Rightarrow \quad \text{Battery Charging} \quad (2.10)$$

Case 2: Balanced Power

When

$$P_G = P_L \quad (2.11)$$

the generated power approximately satisfies the load demand.

Under ideal conditions,

$$P_B \approx 0 \quad (2.12)$$

assuming that other losses and power exchanges are negligible.

Case 3: Insufficient Renewable Generation

When

$$P_G < P_L \quad (2.13)$$

the renewable generation is insufficient to satisfy the load.
The battery therefore supplies the power deficit:

$$P_G - P_L < 0 \Rightarrow \text{Battery Discharging} \quad (2.14)$$

This generation-load relationship forms the basis of the charge-discharge algorithm described in the source material. :contentReference[oaicite:6]index=6

2.5 Electrical Load

The electrical load represents the power consumed by the distributed energy system.
The load power is denoted by

$$P_L \quad (2.15)$$

where P_L represents the instantaneous load demand.

The load determines whether the available renewable generation is sufficient to satisfy the power requirement of the system.

In the power-flow demonstration described in the source material, a **2-kW load** is used. The corresponding simulation investigates how battery charging and discharging compensate for changes in generated power. :contentReference[oaicite:7]index=7

2.6 Battery Energy Storage System

The battery provides the energy-storage capability of the hybrid system.
The battery operates in two fundamental conditions:

2.6.1 Charging Condition

During charging,

$$DC \text{ Bus} \rightarrow Bidirectional \text{ Converter} \rightarrow Battery \quad (2.16)$$

The battery absorbs excess renewable energy.

2.6.2 Discharging Condition

During discharging,

$$Battery \rightarrow Bidirectional \text{ Converter} \rightarrow DC \text{ Bus} \quad (2.17)$$

The battery supplies the power deficit to the DC bus.

Therefore, the battery performs an important buffering function. When renewable generation exceeds the load, the battery absorbs excess energy. When renewable generation falls below the load, the battery supplies the deficit. :contentReference[oaicite:8]index=8

2.6.3 Battery Parameters

The reference simulation uses an idealized battery model with a capacity of

$$Q_I = 2 \text{ Ah} \quad (2.18)$$

where Q_I represents the battery capacity.
The initial charge is

$$Q_0 = 0 \text{ Ah.} \quad (2.19)$$

The charging efficiency is

$$\sigma = 1 \quad (2.20)$$

and the self-discharge is assumed to be zero.

Table 2.1: Reference battery parameters.

Parameter	Symbol	Value
Battery capacity	Q_I	2 Ah
Initial charge	Q_0	0 Ah
Charging efficiency	σ	1
Self-discharge	$S(T_a, t)$	0
Battery SOC	SOC	Variable

These values represent the assumptions used in the reference simulation. If different values are used in the MATLAB/Simulink implementation, the actual simulation parameters should be reported separately.

2.7 Bidirectional DC/DC Converter

The bidirectional DC/DC converter is the **central power-electronic interface** between the battery and the DC bus.

Its principal functions are:

1. Transfer energy from the DC bus to the battery.
2. Transfer energy from the battery to the DC bus.
3. Step down the voltage during charging.
4. Step up the voltage during discharging.
5. Control the battery-side power flow.
6. Support DC-bus voltage regulation.
7. Generate controlled switching through PWM.

The reference system uses a **non-isolated bidirectional DC/DC converter** for charge-discharge control.

2.7.1 Converter Topology

The converter contains two principal semiconductor switches and associated passive components.

The two main switches are denoted by

$$VT_1, \quad VT_2 \quad (2.21)$$

and the associated diodes are denoted by

$$D_1, \quad D_2. \quad (2.22)$$

The principal passive components include the inductor

$$L, \quad (2.23)$$

the capacitors

$$C_1, \quad C_2, \quad (2.24)$$

and the resistive elements

$$R_1, \quad R_2. \quad (2.25)$$

These notations correspond to the converter topology described in the source document. :contentReference[oaicite:11]index=11

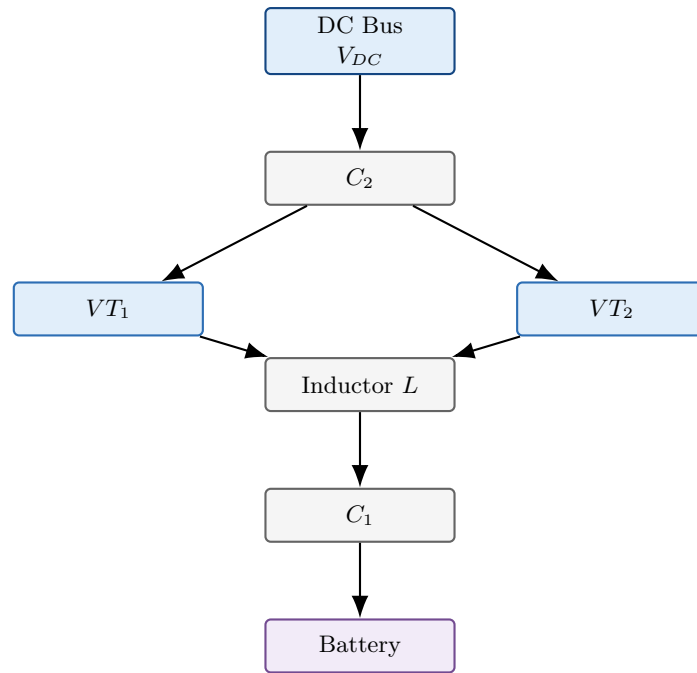


Figure 2.2: Simplified representation of the bidirectional DC/DC converter topology.

Note: Figure 2.2 is a simplified functional representation. The exact circuit topology should be redrawn from the source paper's converter circuit when the final report includes the detailed electrical schematic.

2.7.2 Converter Components and Their Functions

Table 2.2: Main components of the bidirectional DC/DC converter.

Component	Symbol	Function
Main switch 1	VT_1	Controls boost-mode switching
Main switch 2	VT_2	Controls buck-mode switching
Diode 1	D_1	Provides diode conduction path
Diode 2	D_2	Provides complementary conduction path
Inductor	L	Stores and transfers energy
Capacitor 1	C_1	Filtering and energy storage
Capacitor 2	C_2	Filtering and energy storage
Resistance	R_1, R_2	Represents resistive elements

The detailed topology and component arrangement will be analyzed mathematically in the converter-modelling chapter.

2.8 Operating Modes of the Bidirectional Converter

The bidirectional converter has two principal energy-transfer modes: **buck mode** and **boost mode**.

2.8.1 Buck Mode – Battery Charging

During buck operation, power flows from the DC bus toward the battery:

$$\boxed{DC \text{ Bus} \rightarrow Converter \rightarrow Battery} \quad (2.26)$$

The converter therefore operates as a step-down converter.
The switching condition is

$$VT_1 = \text{OFF} \quad (2.27)$$

and

$$VT_2 = \text{ON}. \quad (2.28)$$

For an ideal buck converter, the voltage relationship is

$$\boxed{D = \frac{U_o}{U_i}} \quad (2.29)$$

where

- D is the duty ratio,
- U_i is the converter input voltage, and
- U_o is the desired output voltage.

The detailed buck-mode operation and mathematical derivation are presented in Chapter 3.

2.8.2 Boost Mode – Battery Discharging

During boost operation, power flows from the battery toward the DC bus:

$$\boxed{Battery \rightarrow Converter \rightarrow DC \text{ Bus}} \quad (2.30)$$

The converter therefore operates as a step-up converter.
The switching condition is

$$VT_1 = \text{ON} \quad (2.31)$$

and

$$VT_2 = \text{OFF}. \quad (2.32)$$

For an ideal boost converter,

$$\boxed{D = 1 - \frac{U_i}{U_o}} \quad (2.33)$$

where U_i is the converter input voltage and U_o is the desired output voltage.

The detailed boost-mode derivation will be presented in Chapter 3. The operating conditions described above follow the uploaded source material. :contentReference[oaicite:12]index=12

2.9 Switching-State Configuration

The two main switches VT_1 and VT_2 determine the operating condition of the bidirectional converter.

Table 2.3: Switching states of the bidirectional converter.

State	VT_1	VT_2	Converter Condition
1	OFF	ON	Buck / Step-down
2	ON	OFF	Boost / Step-up
3	OFF	OFF	Both switches OFF
4	ON	ON	Forbidden

The simultaneous ON condition of VT_1 and VT_2 must be prevented. The source material explicitly identifies this state as forbidden and describes the use of additional switching logic to prevent simultaneous conduction. :contentReference[oaicite:13]index=13

2.9.1 Reason for Preventing Simultaneous Conduction

If both switches are simultaneously turned ON, an undesirable conduction path can be established in the converter. This can lead to improper converter operation and potentially excessive current.

Therefore, the control system must ensure that the switching commands satisfy

$$\boxed{VT_1 \neq \text{ON} \quad \text{and} \quad VT_2 \neq \text{ON} \quad \text{simultaneously.}} \quad (2.34)$$

This switching protection forms an important part of the converter control architecture.

2.10 Power-Flow Operating Conditions

The complete hybrid system can be understood through three principal power-flow conditions.

2.10.1 Condition A – Excess Renewable Generation

When

$$P_G > P_L, \quad (2.35)$$

the renewable sources generate more power than the load consumes. The excess power is therefore directed toward the battery.

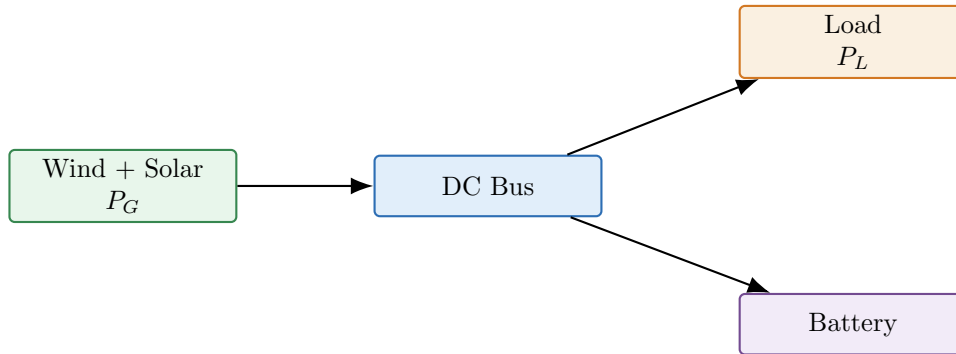


Figure 2.3: Power flow during excess renewable generation, $P_G > P_L$.

Thus,

$$\boxed{P_G - P_L > 0 \Rightarrow \text{Battery Charging.}} \quad (2.36)$$

2.10.2 Condition B – Balanced Power

When

$$P_G \approx P_L, \quad (2.37)$$

the renewable generation approximately satisfies the load demand. The battery power is ideally close to zero:

$$\boxed{P_B \approx 0.} \quad (2.38)$$

2.10.3 Condition C – Insufficient Renewable Generation

When

$$P_G < P_L, \quad (2.39)$$

the renewable sources cannot supply the complete load demand.

The battery therefore supplies the required deficit through the bidirectional converter.



Figure 2.4: Power flow during insufficient renewable generation, $P_G < P_L$.

Therefore,

$$P_G - P_L < 0 \Rightarrow \text{Battery Discharging.} \quad (2.40)$$

These three operating conditions form the basis of the energy management strategy described in the source material. :contentReference[oaicite:14]index=14

2.11 Battery SOC and Energy-Management Interface

The battery cannot be operated solely according to the difference between generated and load power. Its State of Charge (SOC) must also be monitored.

The two important quantities used by the energy-management system are

$$P_G - P_L \quad (2.41)$$

and

$$SOC. \quad (2.42)$$

The simplified decision process is shown in Fig. 2.5.

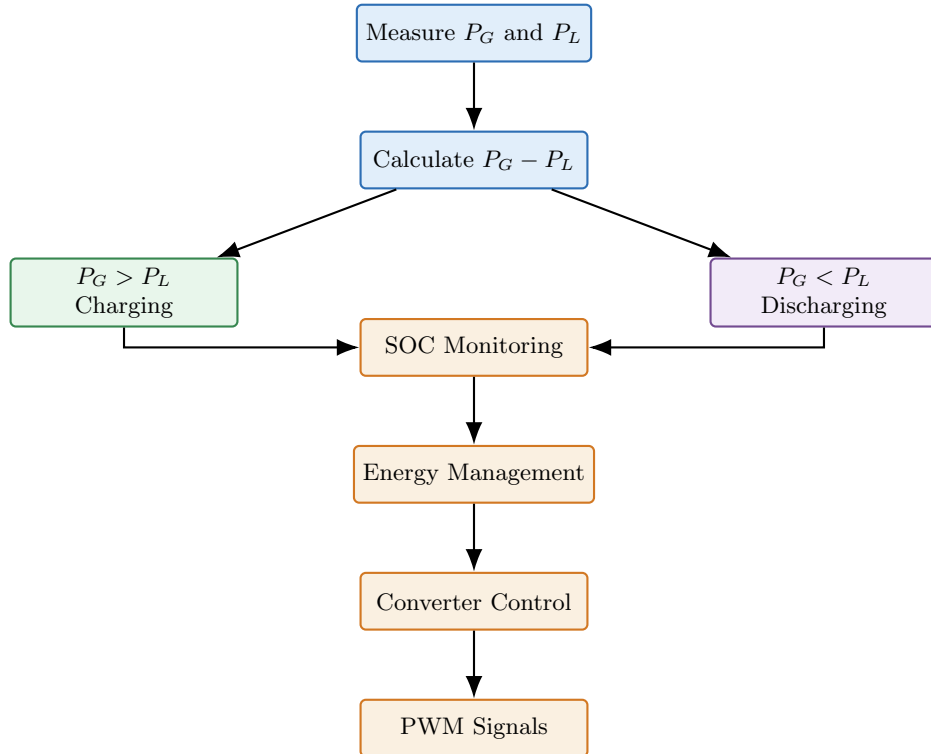


Figure 2.5: Simplified battery SOC and energy-management interface.

The source material uses SOC thresholds of approximately 0.8 and 0.2 for charge/discharge management. :contentReference[oaicite:15]index=15

2.12 Converter Control Interface

The bidirectional converter is operated through a control system that determines the required duty ratio and generates the corresponding PWM switching signals.

The general control interface is

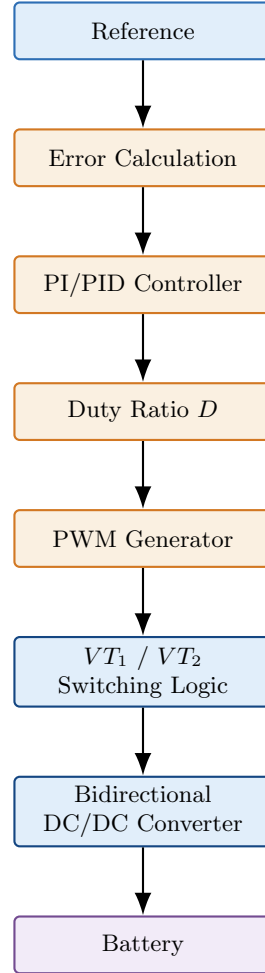


Figure 2.6: Functional interface of the bidirectional converter control system.

The reference implementation uses a triangular carrier of amplitude 1 and a switching frequency of

$$f_{sw} = 20 \text{ kHz.} \quad (2.43)$$

The calculated modulation signal is compared with the carrier to generate the PWM switching commands. :contentReference[oaicite:16]index=16

2.13 Main System Parameters

The principal parameters of the reference system are consolidated in Table 2.4.

Table 2.4: Reference system parameters.

Parameter	Symbol	Reference Value
System rating	P_{rated}	5 kW
DC-bus voltage	V_{DC}	300 V
Switching frequency	f_{sw}	20 kHz
Battery capacity	Q_I	2 Ah
Buck inductance	L_{buck}	350 μH
Boost inductance	L_{boost}	104 μH
Charging efficiency	σ	1
Initial battery charge	Q_0	0 Ah
Self-discharge	$S(T_a, t)$	0
Upper SOC threshold	SOC_{max}	0.8
Lower SOC threshold	SOC_{min}	0.2
Maximum boost duty ratio	D_{max}	0.9

The system rating, DC-bus voltage, switching frequency, battery capacity, converter inductances, SOC thresholds and duty-ratio limit listed above are the reference values provided in the uploaded material.

2.14 Complete Functional Configuration

The complete functional architecture of the system is shown in Fig. 2.7.

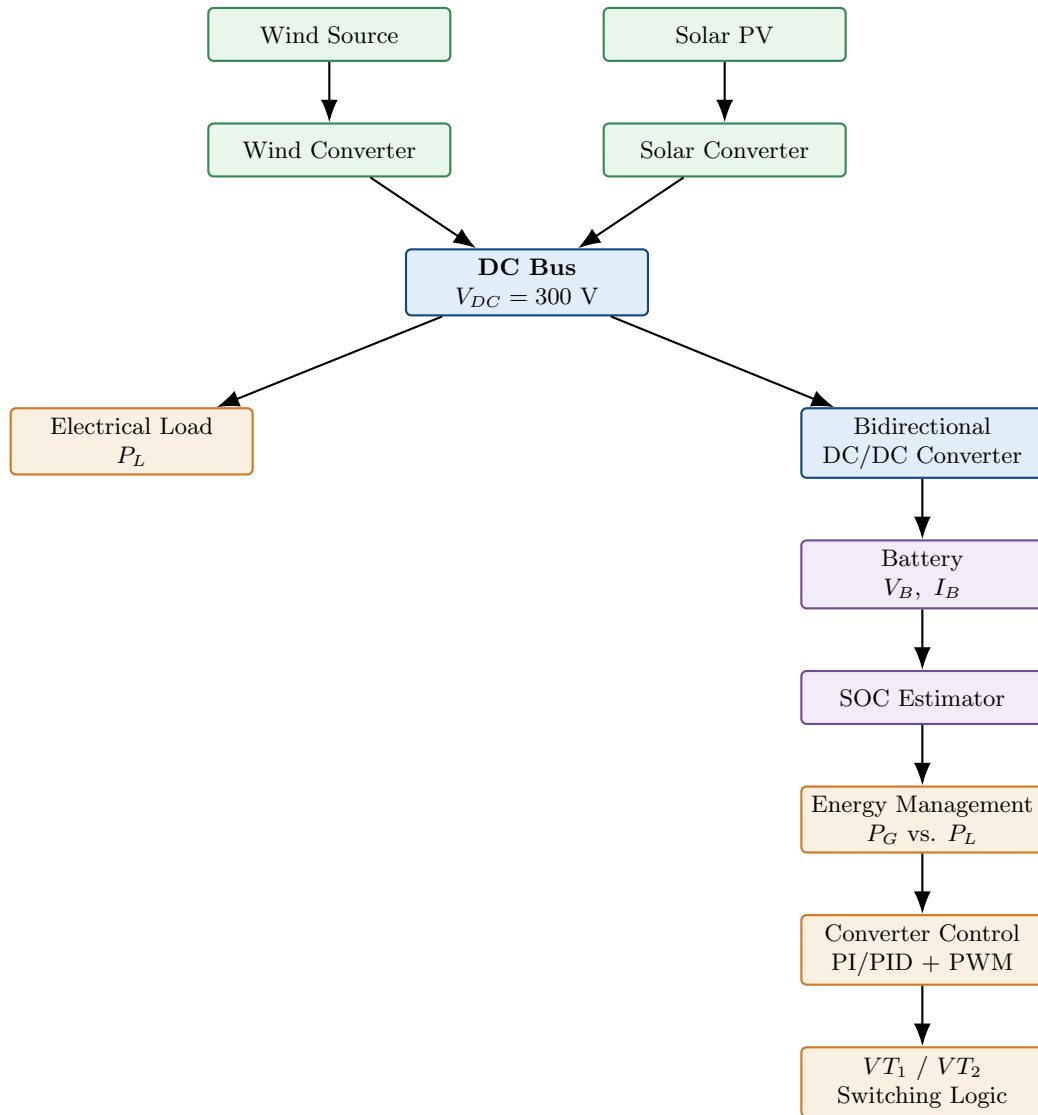


Figure 2.7: Complete functional architecture of the wind-solar hybrid energy-storage system.

This architecture provides the overall relationship between renewable generation, the DC bus, electrical load, bidirectional converter, battery, SOC estimation, energy management and converter control. The complete functional configuration follows the structure developed in the uploaded Chapter 2 material. :contentReference[oaicite:18]index=18

2.15 Notations Introduced in Chapter 2

Table 2.5 summarizes the principal notations introduced in this chapter.

Table 2.5: Notation used in Chapter 2.

Symbol	Description	Unit
P_G	Total renewable generated power	W
P_{wind}	Wind-generated power	W
P_{solar}	Solar-generated power	W
P_L	Load power	W
P_B	Battery power	W
V_{DC}	DC-bus voltage	V
V_B	Battery voltage	V
I_B	Battery current	A
VT_1	Converter switch 1	–
VT_2	Converter switch 2	–
D_1	Converter diode 1	–
D_2	Converter diode 2	–
D	Converter duty ratio	–
L	Converter inductance	H
C_1	Converter capacitor 1	F
C_2	Converter capacitor 2	F
R_1	Converter resistance 1	Ω
R_2	Converter resistance 2	Ω
SOC	Battery State of Charge	% / p.u.
Q_I	Battery capacity	Ah
Q_0	Initial battery charge	Ah
σ	Charging efficiency	–
f_{sw}	Switching frequency	Hz

The notation follows the definitions introduced throughout the uploaded Chapter 2 material. :contentReference[oaicite:19]index=19

Chapter 3

Mathematical Modeling and Design

3.1 Introduction

The previous chapter presented the overall configuration of the 5-kW wind-solar hybrid energy-storage system and described the role of the bidirectional DC/DC converter as the interface between the battery and the DC bus.

This chapter focuses on the mathematical modeling and design of the bidirectional DC/DC converter.

The converter is a non-isolated bidirectional topology that allows electrical energy to flow in two directions:

$$\boxed{\text{Battery} \longrightarrow \text{DC Bus}} \quad (3.1)$$

during battery discharging, and

$$\boxed{\text{DC Bus} \longrightarrow \text{Battery}} \quad (3.2)$$

during battery charging.

The converter topology contains a battery, resistive elements, capacitors, an inductor, two controlled semiconductor switches and two diodes. The switching states of the semiconductor devices determine the operating mode of the converter. :contentReference[oaicite:1]index=1

The mathematical analysis is therefore divided into the following major parts:

1. Detailed converter topology.
2. Component interpretation.
3. Switching-state analysis.
4. Buck-mode operation.
5. Boost-mode operation.
6. Duty-ratio relationships.
7. Inductor design.
8. Converter parameter selection.
9. Comparison of the two operating modes.

3.2 Main Topology of the Bidirectional DC/DC Converter

The bidirectional DC/DC converter used in the system is a **non-isolated converter topology** that serves as the interface between the battery energy-storage system and the DC bus.

Its primary function is to enable controlled transfer of electrical energy in both directions:

$$\boxed{\text{Battery} \rightarrow \text{DC Bus}} \quad (3.3)$$

and

$$\boxed{\text{DC Bus} \rightarrow \text{Battery}} \quad (3.4)$$

The main topology consists of a battery source, series inductance, two controlled semiconductor switches, two diodes, input/output capacitors and resistive/load elements. The reference topology identifies VT_1 and VT_2 as the principal switching devices. Their switching states determine whether the converter operates in step-down or step-up mode. :contentReference[oaicite:2]index=2

3.2.1 Converter Topology

For the final report, the original circuit topology should be redrawn carefully rather than replacing it with an oversimplified circuit.

The major sections of the topology are:

1. Battery-side network.
2. Energy-transfer and switching network.
3. DC-bus/load-side network.

The conceptual organization is

$$\boxed{\begin{array}{ccc} \text{Battery Side} & \rightarrow & \text{Energy Transfer} \\ B, R_1, C_1 & \rightarrow & L, VT_1, D_1, VT_2, D_2 \rightarrow \\ & & \text{DC-Bus Side} \\ & & C_2, R_2, S \end{array}} \quad (3.5)$$

3.2.2 Battery B

The battery is represented by

$$\boxed{B} \quad (3.6)$$

and constitutes the energy-storage element of the system.

The battery can operate in two directions.

During charging,

$$\boxed{DC \text{ Bus} \rightarrow Converter \rightarrow B} \quad (3.7)$$

whereas during discharging,

$$\boxed{B \rightarrow Converter \rightarrow DC \text{ Bus}} \quad (3.8)$$

Thus, the battery is connected to the switching network through the converter inductance and associated passive components.

3.2.3 Battery-Side Resistance R_1

The resistance

$$\boxed{R_1} \quad (3.9)$$

is associated with the battery-side branch of the converter.

It represents the resistive element included in the battery-side electrical model and can be used to account for the resistive characteristics of this branch.

The exact numerical value of R_1 should be taken from the simulation parameters used in the MATLAB/Simulink model. :contentReference[oaicite:4]index=4

3.2.4 Battery-Side Capacitor C_1

The capacitor

$$\boxed{C_1} \quad (3.10)$$

is connected across the battery-side terminals.

Its principal functions are:

- local energy storage,
- voltage filtering,
- reduction of switching-related voltage variations.

Therefore, C_1 can be described as the battery-side capacitor providing local energy storage and filtering of the converter-side voltage.

3.2.5 Inductor L

The inductor

$$\boxed{L} \quad (3.11)$$

is one of the most important components of the bidirectional converter.

It is placed in the main power-transfer path between the battery-side network and the switching network.

The main functions of the inductor are:

1. Energy storage.
2. Energy transfer.
3. Current smoothing.
4. Reduction of current ripple.

The inductor current is represented by

$$\boxed{i_L} \quad (3.12)$$

and constitutes an important state variable in the converter model.

For the reference system, the calculated inductance values are

$$\boxed{L_{\text{buck}} = 350 \mu\text{H}} \quad (3.13)$$

and

$$\boxed{L_{\text{boost}} = 104 \mu\text{H}}. \quad (3.14)$$

These values correspond to the respective buck- and boost-mode design calculations. :contentReference[oaicite:6]index=6

3.2.6 Main Semiconductor Switch VT_1

The first controlled semiconductor device is

$$\boxed{VT_1}. \quad (3.15)$$

The switch is part of the main switching network and is primarily associated with boost operation.

When

$$\boxed{VT_1 = \text{ON}, \quad VT_2 = \text{OFF}} \quad (3.16)$$

the converter operates in step-up mode.

The corresponding power-flow direction is

$$\boxed{Battery \rightarrow Converter \rightarrow DC \text{ Bus}}. \quad (3.17)$$

The reference material identifies this switching condition as the step-up operating condition. :contentReference[oaicite:7]index=7

3.2.7 Diode D_1

The first diode is represented by

$$\boxed{D_1}. \quad (3.18)$$

It is associated with the switching branch containing VT_1 and provides the corresponding diode/freewheeling conduction path during appropriate switching intervals.

Therefore, VT_1 and D_1 should be considered together when analyzing the boost-mode switching states. :contentReference[oaicite:8]index=8

3.2.8 Main Semiconductor Switch VT_2

The second controlled semiconductor device is

$$\boxed{VT_2}. \quad (3.19)$$

It is primarily associated with buck or step-down operation.
When

$$\boxed{VT_1 = \text{OFF}, \quad VT_2 = \text{ON}} \quad (3.20)$$

the converter performs step-down conversion.
The power-flow direction is therefore

$$\boxed{DC \text{ Bus} \rightarrow \text{Converter} \rightarrow \text{Battery}}. \quad (3.21)$$

The reference material identifies this switching condition as the step-down operating condition. :contentReference[oaicite:9]index=9

3.2.9 Diode D_2

The second diode is represented by

$$\boxed{D_2}. \quad (3.22)$$

It is associated with the VT_2 switching branch and provides the corresponding current-conduction path during suitable switching intervals.

The exact diode orientation should be preserved when the original topology is redrawn in the final report. :contentReference[oaicite:10]index=10

3.2.10 DC-Bus Capacitor C_2

The DC-bus-side capacitor is represented by

$$\boxed{C_2}. \quad (3.23)$$

It provides filtering and energy buffering at the DC-bus side of the converter.
The DC-bus voltage is represented by

$$\boxed{V_{DC}} \quad (3.24)$$

and the reference system operates around

$$\boxed{V_{DC} = 300 \text{ V}}. \quad (3.25)$$

The DC-bus capacitor therefore contributes to maintaining a stable DC-bus voltage during converter switching and power-flow transitions. :contentReference[oaicite:11]index=11

3.2.11 Load Resistance R_2

The resistance

$$\boxed{R_2} \quad (3.26)$$

is located on the DC-bus/load side of the converter and represents the resistive load element. The load power is denoted by

$$\boxed{P_L}. \quad (3.27)$$

If R_2 represents a purely resistive load, then

$$\boxed{P_L = \frac{V_{DC}^2}{R_2}}. \quad (3.28)$$

This relationship should be used only when the implemented MATLAB/Simulink load is represented as a purely resistive load. :contentReference[oaicite:12]index=12

3.2.12 Load Switch S

The circuit contains a switch represented by

$$\boxed{S}. \quad (3.29)$$

The switch is shown in series with the resistive load element R_2 .

It can therefore be interpreted as the switching element used to connect or disconnect the load branch.

The exact physical function of S should not be assigned beyond this interpretation unless it is explicitly defined by the original circuit description or the MATLAB/Simulink implementation. :contentReference[oaicite:13]index=13

3.3 Functional Interpretation of the Topology

The complete converter can be understood as an **energy-transfer stage controlled by the semiconductor switches**.

The physical topology determines the switching states, and the switching states determine the converter operating mode.

The overall relationship is

$$\boxed{\text{Physical Topology} \rightarrow \text{Switching State} \rightarrow \text{Operating Mode} \rightarrow \text{Power Flow}} \quad (3.30)$$

3.3.1 Charging Operation

When excess renewable power is available,

$$P_G > P_L, \quad (3.31)$$

the excess energy is transferred from the DC bus to the battery.

The converter therefore operates in buck mode.

The corresponding energy flow is

$$\boxed{DC \text{ Bus} \rightarrow C_2 \rightarrow VT_2/D_2 \rightarrow L \rightarrow C_1 \rightarrow Battery}. \quad (3.32)$$

The corresponding switching condition is

$$VT_1 = \text{OFF}, \quad VT_2 = \text{ON}. \quad (3.33)$$

3.3.2 Discharging Operation

When

$$P_G < P_L, \quad (3.34)$$

the renewable sources cannot satisfy the complete load demand.

The battery therefore supplies the required power to the DC bus.

The converter operates in boost mode.

The corresponding energy flow can be represented as

$$\boxed{Battery \rightarrow C_1 \rightarrow L \rightarrow VT_1/D_1 \rightarrow C_2 \rightarrow DC \text{ Bus}}. \quad (3.35)$$

The switching condition is

$$VT_1 = \text{ON}, \quad VT_2 = \text{OFF}. \quad (3.36)$$

3.4 Switching-State Analysis

The two controlled switches VT_1 and VT_2 determine the operating condition of the converter.

The four possible combinations are summarized in Table 3.1.

Table 3.1: Switching states of the bidirectional DC/DC converter.

State	VT_1	VT_2	Operating Condition
1	OFF	ON	Buck / Step-down
2	ON	OFF	Boost / Step-up
3	OFF	OFF	Inactive / Freewheeling
4	ON	ON	Forbidden

The simultaneous ON state must be prevented because it corresponds to an undesired switching condition. The reference material specifically identifies simultaneous conduction of VT_1 and VT_2 as forbidden. :contentReference[oaicite:14]index=14

3.5 Bidirectional Power Transfer

The term **bidirectional** refers to the capability of the converter to transfer electrical energy in both directions.

3.5.1 DC Bus to Battery

During charging,

$$\boxed{DC \text{ Bus} \rightarrow Battery} \quad (3.37)$$

and the converter performs buck operation.

3.5.2 Battery to DC Bus

During discharging,

$$\boxed{Battery \rightarrow DC \text{ Bus}} \quad (3.38)$$

and the converter performs boost operation.

Therefore,

$$\boxed{\text{Bidirectional Converter} = \text{Buck Operation} + \text{Boost Operation}}. \quad (3.39)$$

This bidirectional energy-transfer capability is the fundamental feature of the converter used in the hybrid energy-storage system. :contentReference[oaicite:15]index=15

3.6 Power Flow and Control Flow

It is important to distinguish between the electrical power flow and the control-information flow.

3.6.1 Electrical Power Flow

The electrical power flow can be represented as

$$\boxed{\text{Renewable Sources} \rightarrow \text{DC Bus} \leftrightarrow \text{Bidirectional Converter} \leftrightarrow \text{Battery}} \quad (3.40)$$

where the bidirectional arrows represent charging and discharging operation.

3.6.2 Control Flow

The control-information flow can be represented as

$$\boxed{\text{Measurements} \rightarrow \text{Power/SOC Information} \rightarrow \text{Energy Management} \rightarrow \text{Controller} \rightarrow \text{PWM} \rightarrow VT_1/VT_2} \quad (3.41)$$

The electrical power flow carries energy, whereas the control flow carries measurement and control information. :contentReference[oaicite:16]index=16

3.7 Relation Between Topology and Mathematical Model

The physical topology directly determines the mathematical models used for the two operating modes.

$$\begin{array}{c} \boxed{\begin{array}{c} \text{Physical Topology} \\ \downarrow \\ \text{Switching State} \end{array}} \Rightarrow \boxed{\begin{array}{c} VT_1 = \text{OFF} \\ VT_2 = \text{ON} \end{array}} \Rightarrow \boxed{\text{Buck Mode}} \end{array} \quad (3.42)$$

Similarly,

$$\boxed{\begin{array}{c} VT_1 = \text{ON} \\ VT_2 = \text{OFF} \end{array}} \Rightarrow \boxed{\text{Boost Mode}}. \quad (3.43)$$

For ideal converter operation, the corresponding voltage-duty relationships are

$$\boxed{D = \frac{U_o}{U_i}} \quad (\text{Buck}) \quad (3.44)$$

and

$$\boxed{D = 1 - \frac{U_i}{U_o}} \quad (\text{Boost}). \quad (3.45)$$

These relationships provide the basis for the detailed mathematical derivation presented in the subsequent sections.

3.8 Parameter Summary of the Converter Topology

The main components and variables introduced in this chapter are summarized in Table 3.2.

Table 3.2: Principal converter components and their functions.

Symbol	Component	Main Function
B	Battery	Energy storage
R_1	Battery-side resistance	Represents resistive element
C_1	Battery-side capacitor	Filtering and energy buffering
L	Inductor	Energy storage and current smoothing
VT_1	Controlled switch	Primarily associated with boost operation
D_1	Diode	Complementary conduction path for VT_1
VT_2	Controlled switch	Primarily associated with buck operation
D_2	Diode	Complementary conduction path for VT_2
C_2	DC-bus capacitor	DC-bus filtering and energy buffering
S	Load switch	Load connection/disconnection
R_2	Load resistance	Represents resistive load
V_{DC}	DC-bus voltage	Common DC voltage

Chapter 4

Control Strategy and PWM Implementation

4.1 Introduction

The bidirectional DC/DC converter developed in Chapter 3 provides the required electrical path for energy transfer between the battery and the DC bus. However, the converter cannot operate properly without an appropriate control system. The purpose of the control system is to determine the required switching duty ratio and generate the corresponding gate pulses for the semiconductor switches.

The control strategy must perform two different tasks depending on the direction of power flow:

$$\boxed{\text{DC Bus} \rightarrow \text{Battery} \quad \Rightarrow \quad \text{Buck Control}} \quad (4.1)$$

and

$$\boxed{\text{Battery} \rightarrow \text{DC Bus} \quad \Rightarrow \quad \text{Boost Control}} \quad (4.2)$$

The reference paper develops separate control relationships for the buck and boost operating conditions and implements them using MATLAB/Simulink. A triangular carrier with amplitude 1 and switching frequency of 20 kHz is used to generate the PWM signals.

The complete control architecture can be represented as follows:

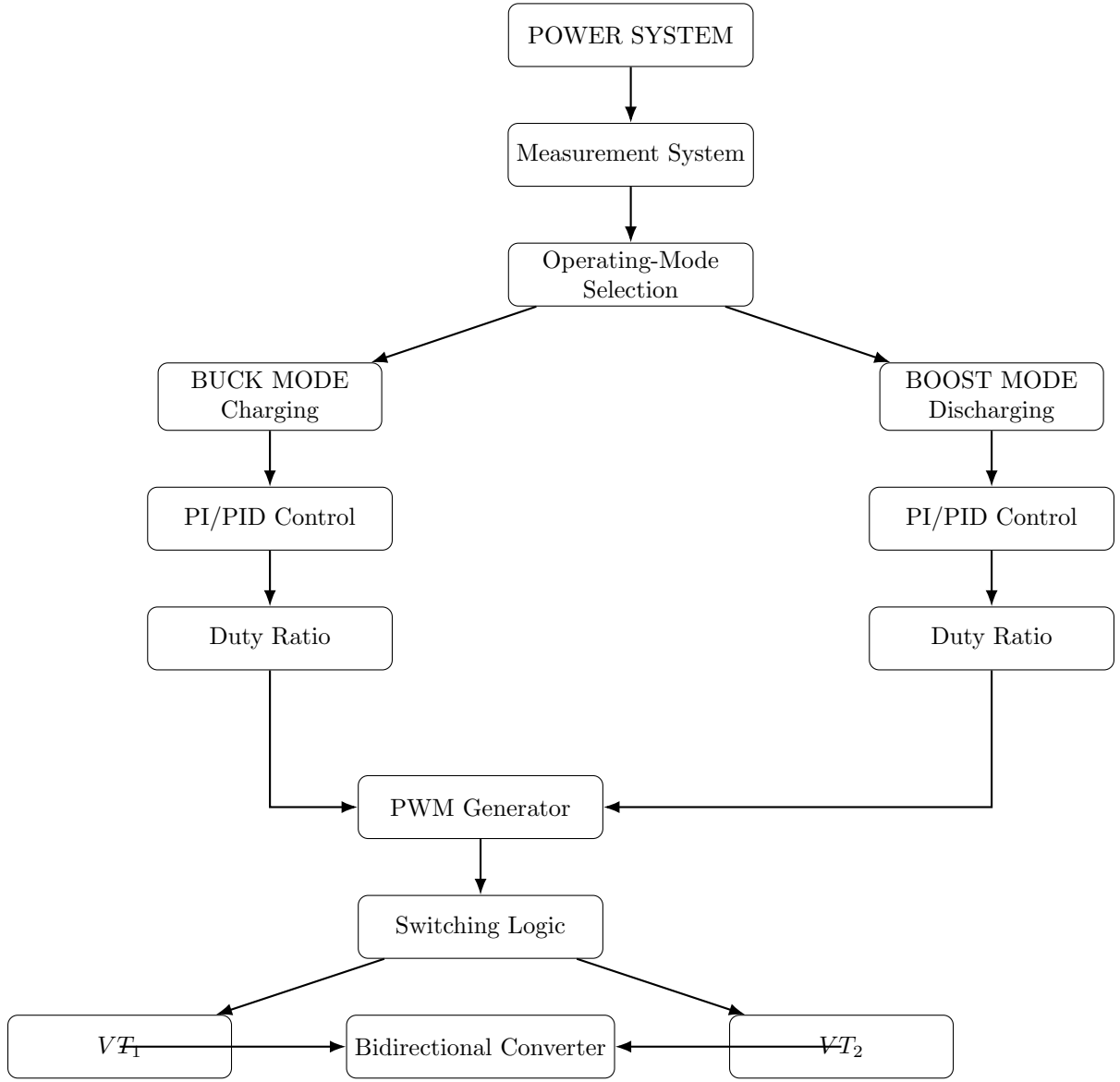


Figure 4.1: Overall control architecture of the bidirectional DC/DC converter.

4.2 Control Objectives

The converter control system has several objectives.

4.2.1 Objective 1 — Voltage Regulation

The converter should regulate the relevant output voltage around its reference:

$$V_o(t) \rightarrow V_o^* \quad (4.3)$$

where V_o^* is the desired voltage.

4.2.2 Objective 2 — Controlled Power Transfer

The converter must allow controlled energy transfer:

$$\text{DC Bus} \rightarrow \text{Battery} \quad (4.4)$$

during charging and

$$\text{Battery} \rightarrow \text{DC Bus} \quad (4.5)$$

during discharging.

4.2.3 Objective 3 — Appropriate Duty-Ratio Generation

The controller must determine a suitable duty ratio

$$\boxed{D(t)} \tag{4.6}$$

for the corresponding operating condition.

4.2.4 Objective 4 — Safe Switching

The controller must prevent the two main switches

$$VT_1 \tag{4.7}$$

and

$$VT_2 \tag{4.8}$$

from being simultaneously ON.

The switching logic is therefore introduced to avoid this condition.

4.3 Overall Control Architecture

The control system can be divided into five major stages:

1. Measurement
2. Error Calculation
3. PI/PID Control
4. PWM Generation
5. Switching Logic

These stages are interconnected with the power converter and feedback measurements.

4.3.1 General Converter Control Architecture

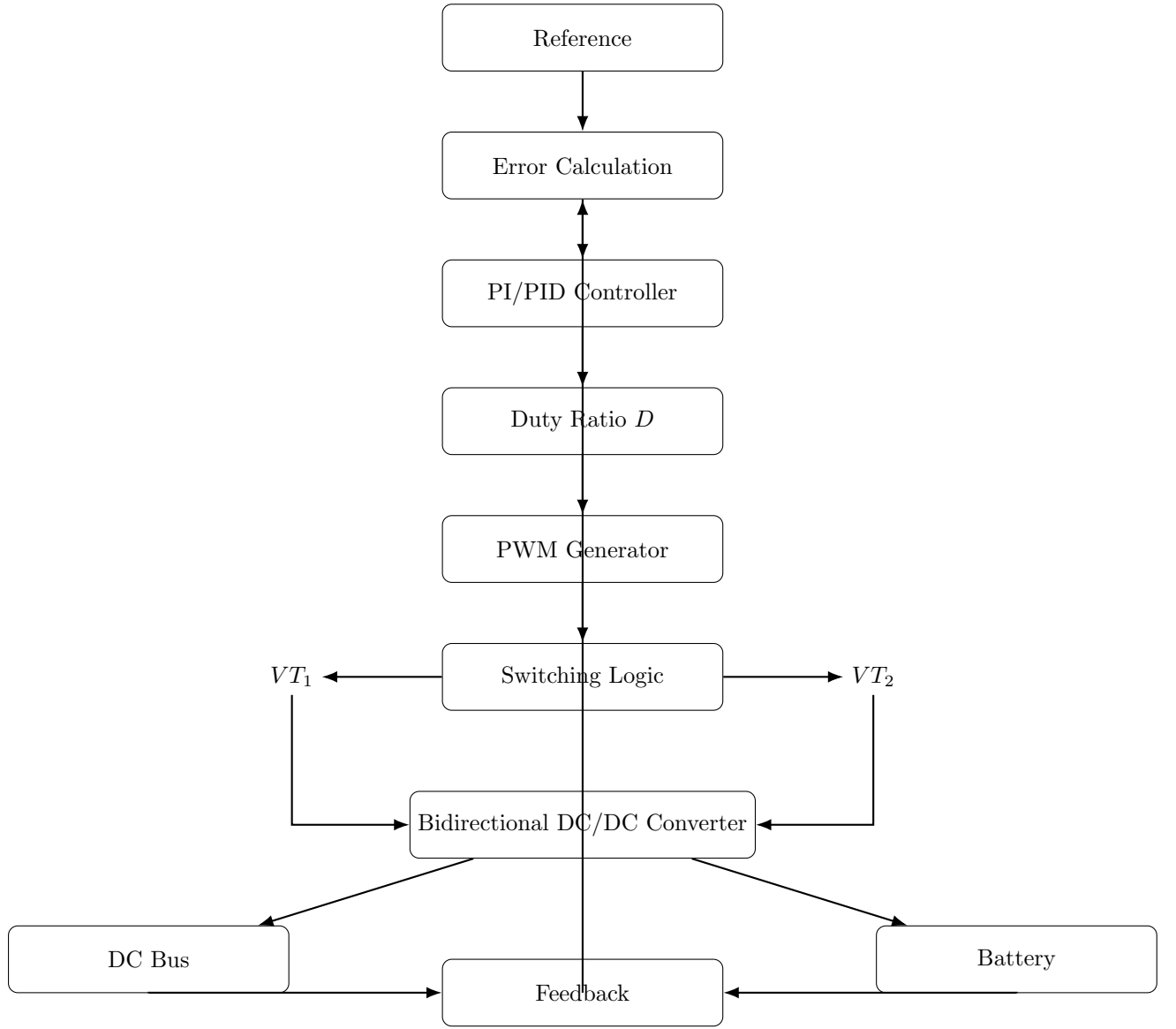


Figure 4.2: General control architecture of the bidirectional DC/DC converter.

4.4 Operating-Mode Selection

Before the appropriate controller is activated, the direction of energy transfer must be determined.

The basic system-level criterion is:

$$P_G > P_L \Rightarrow \text{Charging} \quad (4.9)$$

and

$$P_G < P_L \Rightarrow \text{Discharging} \quad (4.10)$$

where

$$P_G = \text{renewable generated power} \quad (4.11)$$

and

$$P_L = \text{load power}. \quad (4.12)$$

Therefore,

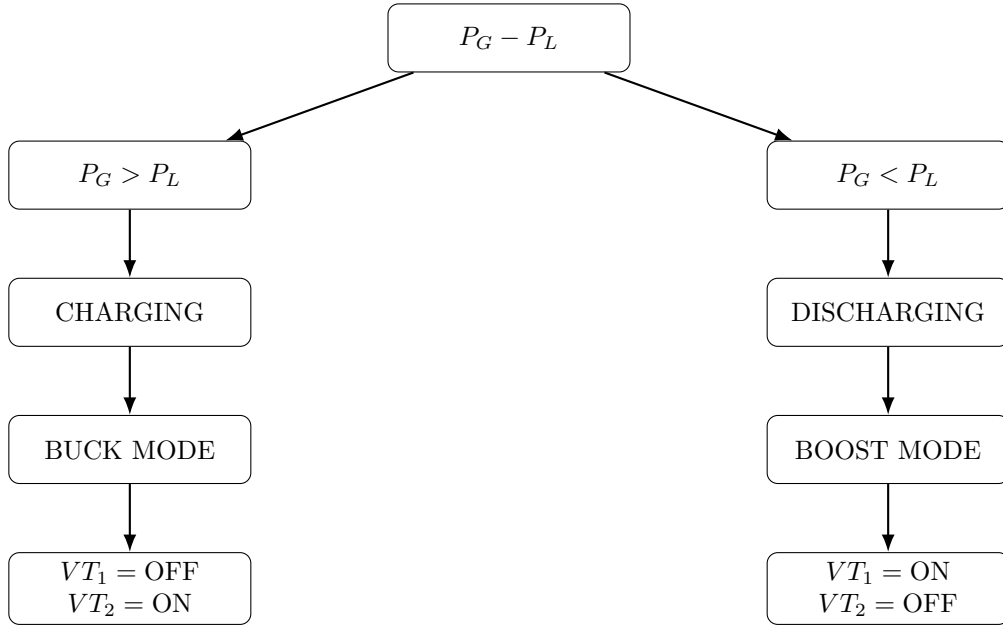


Figure 4.3: Operating-mode selection based on the power imbalance.

The charge/discharge decision is used together with the SOC limits.

4.5 Buck-Mode Control

4.5.1 Operating Condition

Buck operation occurs when the battery needs to absorb energy from the DC bus.

Therefore,

$$\boxed{P_G > P_L} \quad (4.13)$$

generally corresponds to charging.

The power-flow direction is

$$\boxed{\text{DC Bus} \rightarrow \text{Converter} \rightarrow \text{Battery}} \quad (4.14)$$

The corresponding switch condition is

$$\boxed{VT_1 = \text{OFF}, \quad VT_2 = \text{ON}} \quad (4.15)$$

which represents the step-down operating state.

4.5.2 Buck Voltage Relationship

From Chapter 3, the ideal buck relationship is

$$\boxed{D = \frac{U_o}{U_i}} \quad (4.16)$$

Therefore,

$$\boxed{U_o = DU_i} \quad (4.17)$$

This means that the output voltage can be regulated by adjusting the duty ratio.

4.5.3 Buck Voltage Error

The voltage error is defined as

$$e(t) = U_o(t) - U'_o(t) \quad (4.1)$$

where:

- $U_o(t)$ is the desired output voltage,
- $U'_o(t)$ is the actual output voltage,
- $e(t)$ is the voltage error.

The control objective is

$$e(t) \rightarrow 0 \quad (4.18)$$

4.5.4 Buck PI/PID Controller

The error signal is supplied to the controller.

Conceptually,

$$e(t) \rightarrow PI/PID \rightarrow \text{Control Correction.} \quad (4.19)$$

For a PI controller, the general control law is

$$u(t) = K_P e(t) + K_I \int e(t) dt \quad (4.2)$$

where:

- K_P is the proportional gain,
- K_I is the integral gain.

If a PID controller is used, the general form is

$$u(t) = K_P e(t) + K_I \int e(t) dt + K_D \frac{de(t)}{dt} \quad (4.3)$$

where K_D is the derivative gain.

Note: Use PI or PID according to the actual controller implemented in the Simulink model.

4.5.5 Buck Duty-Ratio Generation

The controlled duty ratio is expressed as

$$D(t) = \frac{U_o(t) + PI[e(t)]}{U_i(t)} \quad (4.4)$$

This equation combines the nominal duty ratio with the controller correction.

The complete process is:

$$\text{Desired } U_o \rightarrow \text{Error Calculation} \rightarrow e(t) \rightarrow PI/PID \rightarrow \text{Duty-Ratio Correction} \rightarrow \text{PWM} \rightarrow VT_2 \quad (4.20)$$

4.6 Boost-Mode Control

4.6.1 Operating Condition

Boost operation occurs when the renewable sources cannot supply the required load power.

Therefore,

$$\boxed{P_G < P_L} \quad (4.21)$$

The battery supplies the power deficit:

$$\boxed{\text{Battery} \rightarrow \text{Converter} \rightarrow \text{DC Bus}} \quad (4.22)$$

The corresponding switch condition is

$$\boxed{VT_1 = \text{ON}, \quad VT_2 = \text{OFF}} \quad (4.23)$$

which identifies the step-up operating mode.

4.6.2 Boost Voltage Relationship

The ideal boost relationship is

$$\boxed{D = 1 - \frac{U_i}{U_o}} \quad (4.5)$$

Equivalently,

$$\boxed{U_o = \frac{U_i}{1 - D}} \quad (4.6)$$

Thus, increasing the duty ratio increases the ideal voltage conversion ratio.

4.6.3 Desired and Actual Duty Ratio

The desired duty ratio is

$$\boxed{D(t) = 1 - \frac{U_i(t)}{U_o(t)}} \quad (4.7)$$

and the actual duty ratio is

$$\boxed{D'(t) = 1 - \frac{U_i(t)}{U'_o(t)}} \quad (4.8)$$

The distinction between desired and actual quantities allows the control system to determine the required correction.

4.6.4 Boost Control Error

The error is based on the difference between desired and actual duty ratios:

$$\boxed{e(t) = D(t) - D'(t)} \quad (4.9)$$

This error is then supplied to the PI/PID controller.

The control objective is

$$\boxed{e(t) \rightarrow 0} \quad (4.24)$$

4.6.5 Boost Duty-Ratio Control

The controlled duty-ratio expression is

$$D(t) = 1 - \frac{U_i(t)}{U_o(t)} + PI[e(t)] \quad (4.10)$$

The first term provides the nominal duty ratio while the controller provides the correction required to compensate for the deviation from the desired operating condition.

4.6.6 Boost Duty-Ratio Limitation

The duty ratio is limited to

$$0 \leq D \leq 0.9 \quad (4.11)$$

This limitation is important because an unrestricted duty ratio can lead to undesirable converter behaviour.

A practical Simulink implementation can be represented as:

$$\text{Controller Output} \rightarrow \text{Saturation } (0 \leq D \leq 0.9) \rightarrow \text{PWM Generator} \quad (4.25)$$

4.7 PWM Generation

The controller does not directly switch the semiconductor devices. Instead, the duty ratio is converted into a high-frequency PWM signal.

The PWM process consists of:

1. The controller generates D .
2. D is compared with a carrier.
3. The comparison produces a pulse.
4. The pulse controls the corresponding semiconductor switch.

The triangular carrier has amplitude

$$A_c = 1 \quad (4.26)$$

and switching frequency

$$f_{sw} = 20 \text{ kHz} \quad (4.27)$$

4.7.1 PWM Principle

The PWM principle can be represented using a comparison between the control signal and the triangular carrier.

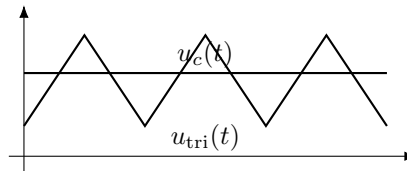


Figure 4.4: Conceptual PWM generation using a triangular carrier.

Mathematically, the switching signal can be expressed as

$$PWM(t) = \begin{cases} 1, & u_c(t) > u_{tri}(t), \\ 0, & u_c(t) \leq u_{tri}(t). \end{cases} \quad (4.28)$$

where:

- $u_c(t)$ is the controller/modulating signal,
- $u_{tri}(t)$ is the triangular carrier.

4.8 Switching Frequency

The switching frequency is

$$\boxed{f_{sw} = 20 \text{ kHz}} \quad (4.29)$$

Therefore, the switching period is

$$T_{sw} = \frac{1}{f_{sw}} \quad (4.30)$$

and hence

$$\boxed{T_{sw} = 50 \text{ } \mu\text{s}} \quad (4.12)$$

The switching frequency affects:

- Current ripple,
- Voltage ripple,
- Inductor size,
- Switching losses,
- Dynamic response.

4.9 Switching Logic

The PWM signals must be passed through switching logic before being applied to VT_1 and VT_2 .

The principal requirement is

$$\boxed{VT_1 \neq \text{ON} \quad \text{when} \quad VT_2 = \text{ON}} \quad (4.31)$$

and vice versa.

The valid switching states are summarized in Table 4.1.

Table 4.1: Valid switching states of the bidirectional converter.

VT_1	VT_2	Mode
OFF	ON	Buck
ON	OFF	Boost
OFF	OFF	Allowed inactive state
ON	ON	Forbidden

Additional switching logic is introduced to prevent the simultaneous ON state.

4.10 Complementary Switching Logic

The switching logic can be represented as:

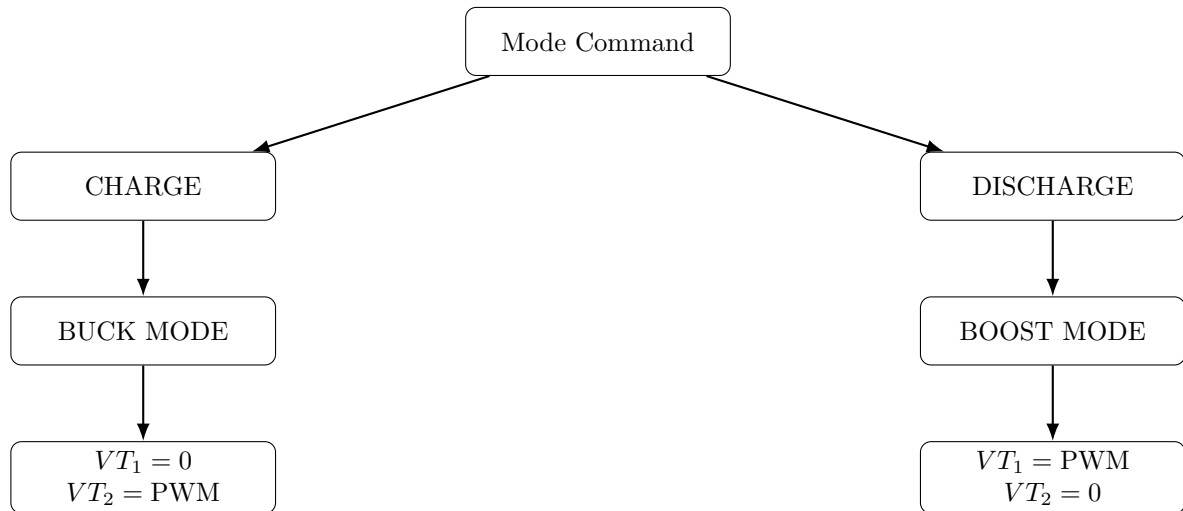


Figure 4.5: Complementary switching logic for buck and boost operation.

In a practical implementation, dead time may also be introduced between complementary switch commands. If dead time is not actually implemented in the Simulink model, it should not be claimed as an implemented feature.

4.11 Integrated Control Architecture

The complete controller can now be represented as:

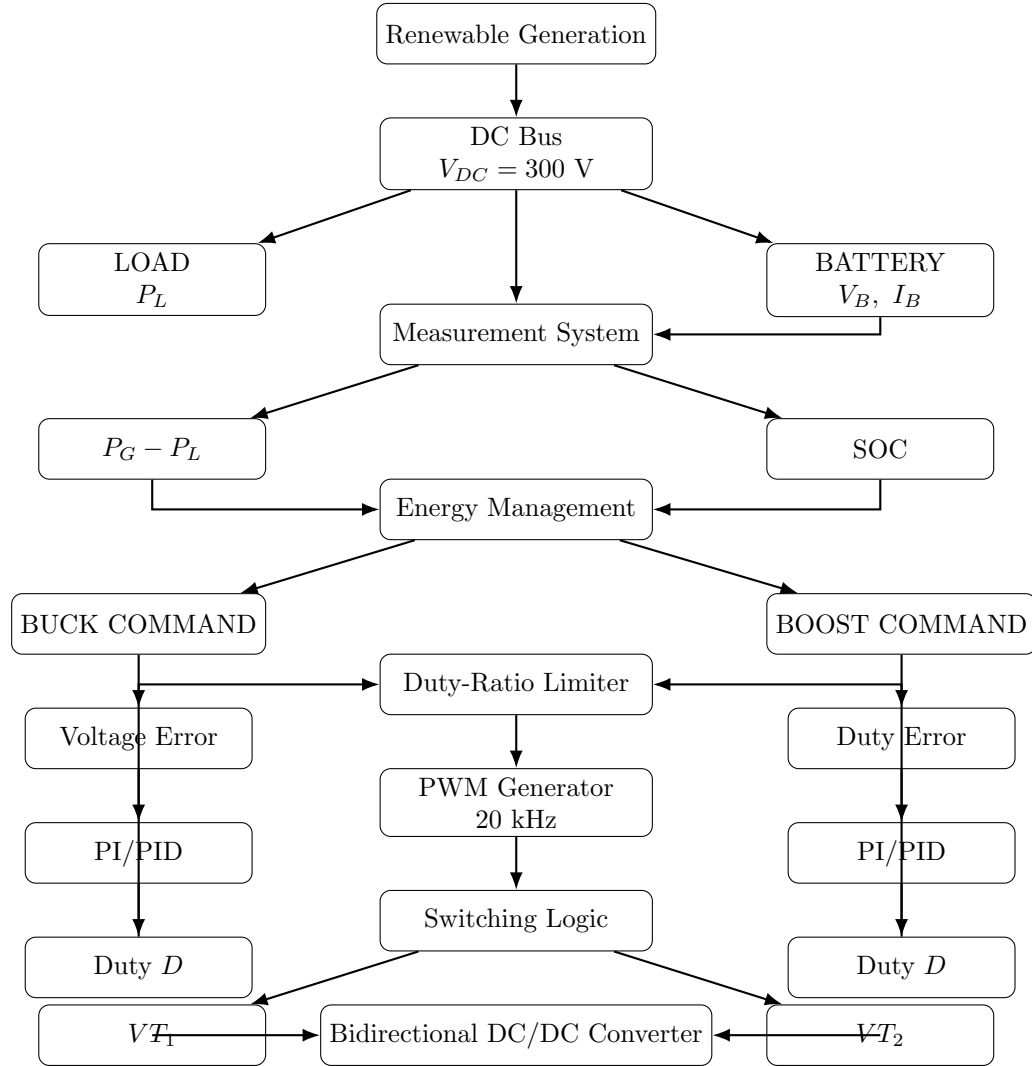


Figure 4.6: Integrated control architecture of the bidirectional DC/DC converter.

This is the main control diagram for Chapter 4.

4.12 Control Sequence During Charging

The charging control sequence is:

Step 1

Measure renewable generation and load:

$$P_G, \quad P_L. \quad (4.32)$$

Step 2

Determine

$$P_G > P_L. \quad (4.33)$$

Step 3

Select

$$\boxed{\text{Buck Mode}} \quad (4.34)$$

Step 4

Set

$$VT_1 = \text{OFF} \quad (4.35)$$

and

$$VT_2 = \text{ON/PWM}. \quad (4.36)$$

Step 5

Calculate the voltage error:

$$e(t) = U_o(t) - U'_o(t). \quad (4.37)$$

Step 6

Apply PI/PID control.

Step 7

Generate the duty ratio.

Step 8

Generate PWM.

Step 9

Transfer energy:

$$\boxed{\text{DC Bus} \rightarrow \text{Battery}} \quad (4.38)$$

4.13 Control Sequence During Discharging

The discharging control sequence is:

Step 1

Measure

$$P_G, \quad P_L. \quad (4.39)$$

Step 2

Determine

$$P_G < P_L. \quad (4.40)$$

Step 3

Select

$$\boxed{\text{Boost Mode}} \quad (4.41)$$

Step 4

Set

$$VT_1 = \text{ON/PWM} \quad (4.42)$$

and

$$VT_2 = \text{OFF}. \quad (4.43)$$

Step 5

Calculate the desired duty ratio:

$$D = 1 - \frac{U_i}{U_o}. \quad (4.44)$$

Step 6

Calculate the control error.

Step 7

Apply PI/PID correction.

Step 8

Limit the duty ratio:

$$\boxed{0 \leq D \leq 0.9} \quad (4.45)$$

Step 9

Generate PWM.

Step 10

Transfer energy:

$$\boxed{\text{Battery} \rightarrow \text{DC Bus}} \quad (4.46)$$

4.14 Interaction Between SOC and Converter Control

The converter controller should not operate independently of the battery-management system.

The SOC provides a constraint on the allowed energy-transfer direction.

The simplified logic is:

$$SOC > 0.8 \quad (4.47)$$

and

$$SOC < 0.2. \quad (4.48)$$

Therefore, the final control decision should conceptually depend on both

$$\boxed{P_G - P_L} \quad (4.49)$$

and

$$\boxed{SOC} \quad (4.50)$$

which gives

$$\boxed{\text{Energy Management} = f(P_G, P_L, SOC)} \quad (4.51)$$

rather than relying only on the renewable/load power difference.

4.15 MATLAB/Simulink Implementation

The controller can be implemented in MATLAB/Simulink using the following functional blocks:

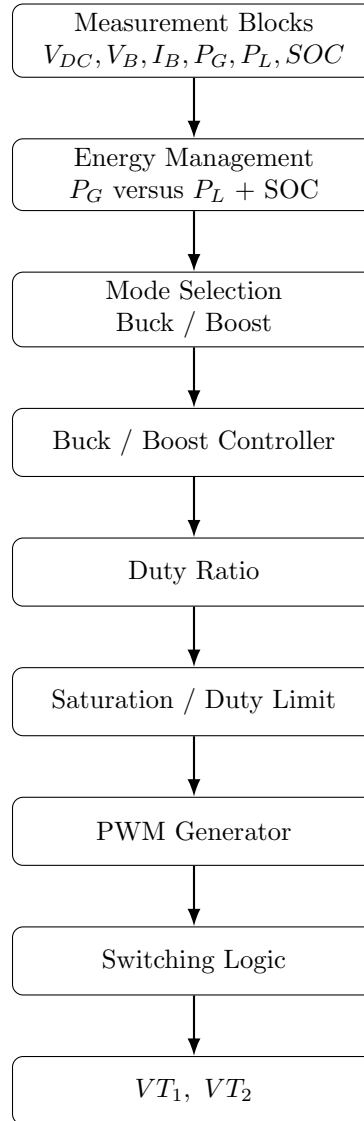


Figure 4.7: Functional structure for MATLAB/Simulink implementation.

This modular implementation is preferable because each block has a clearly defined function.

4.16 Control Parameters

The following table summarizes the control parameters specified in the reference implementation.

Table 4.2: Control parameters of the converter.

Parameter	Symbol	Value
DC-bus reference	V_{DC}^*	300 V
Switching frequency	f_{sw}	20 kHz
Carrier amplitude	A_c	1
Maximum boost duty	$D_{\max}$	0.9
Minimum duty	$D_{\min}$	0
Proportional gain	K_P	<i>Your value</i>
Integral gain	K_I	<i>Your value</i>
Derivative gain	K_D	<i>If applicable</i>
Upper SOC limit	$SOC_{\max}$	0.8
Lower SOC limit	$SOC_{\min}$	0.2

Important: Do not insert fabricated values for K_P , K_I , or K_D . These must be replaced by the values from your actual controller implementation.

4.17 Control Performance Indicators

The controller performance should eventually be evaluated using measurable quantities.

For DC-bus regulation,

$$e_v(t) = V_{DC}^* - V_{DC}(t). \quad (4.52)$$

4.17.1 Maximum Error

$$e_{\max} = \max |e_v(t)| \quad (4.53)$$

4.17.2 Mean Absolute Error

$$MAE = \frac{1}{T} \int_0^T |e_v(t)| dt \quad (4.54)$$

4.17.3 Settling Time

$$T_s = \text{time required for the response to remain within the specified tolerance band} \quad (4.55)$$

These metrics can be used in the results chapter to quantitatively evaluate the controller.

Chapter 4 Root Map

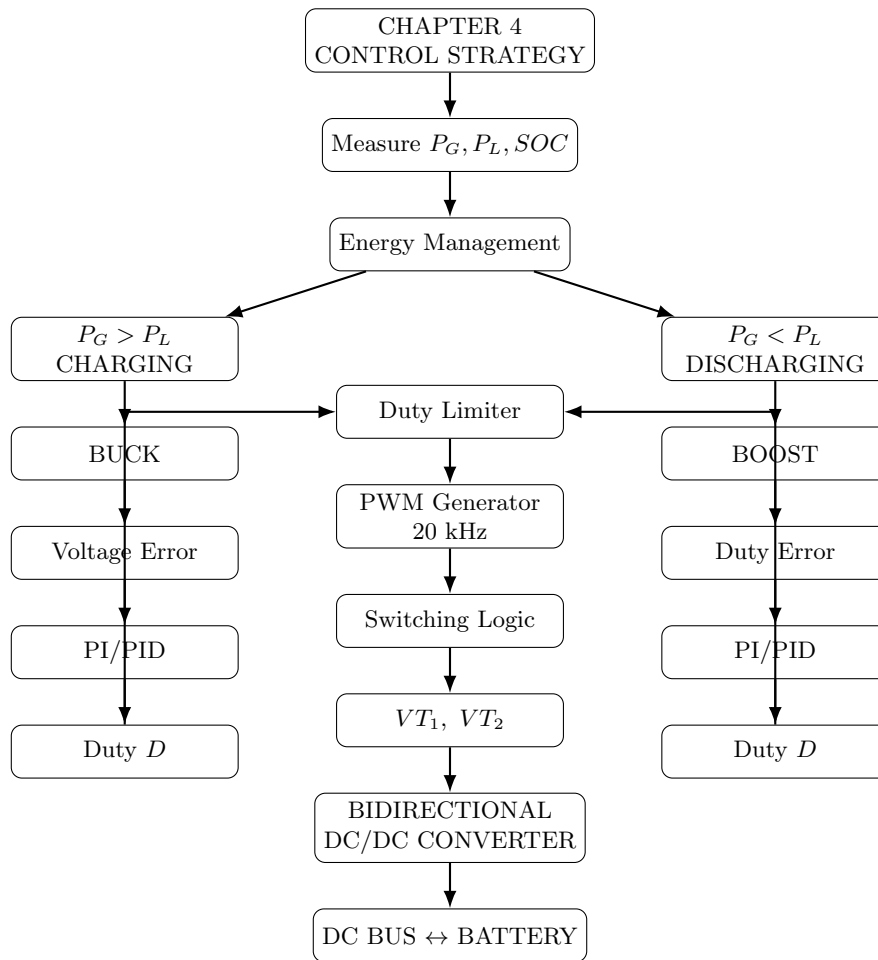


Figure 4.8: Root map of the Chapter 4 control strategy.

Chapter 5

Battery Model, SOC Estimation and Energy Management

5.1 Introduction

The battery energy-storage system is a critical subsystem of the 5-kW wind-solar hybrid system because it compensates for the mismatch between renewable power generation and load demand. Renewable generation and load are both time-varying; therefore, the battery must alternately absorb excess energy and supply energy when generation is insufficient.

The battery-management section of this work has three closely connected functions:

$$\boxed{\text{Power-Flow Detection} \rightarrow \text{SOC Estimation} \rightarrow \text{Charge/Discharge Decision}} \quad (5.1)$$

The battery State of Charge (SOC) is estimated using the **Ampere-Hour method**, while the charge/discharge scheduling in the reference work is described using an **SOC-based dynamic-programming approach**.

The resulting decision is then passed to the bidirectional DC/DC converter, which performs the required buck or boost operation.

5.2 Role of the Battery in the Hybrid System

The battery is connected to the DC bus through the bidirectional DC/DC converter.

The overall energy-storage interface is illustrated in Figure [5.1](#).

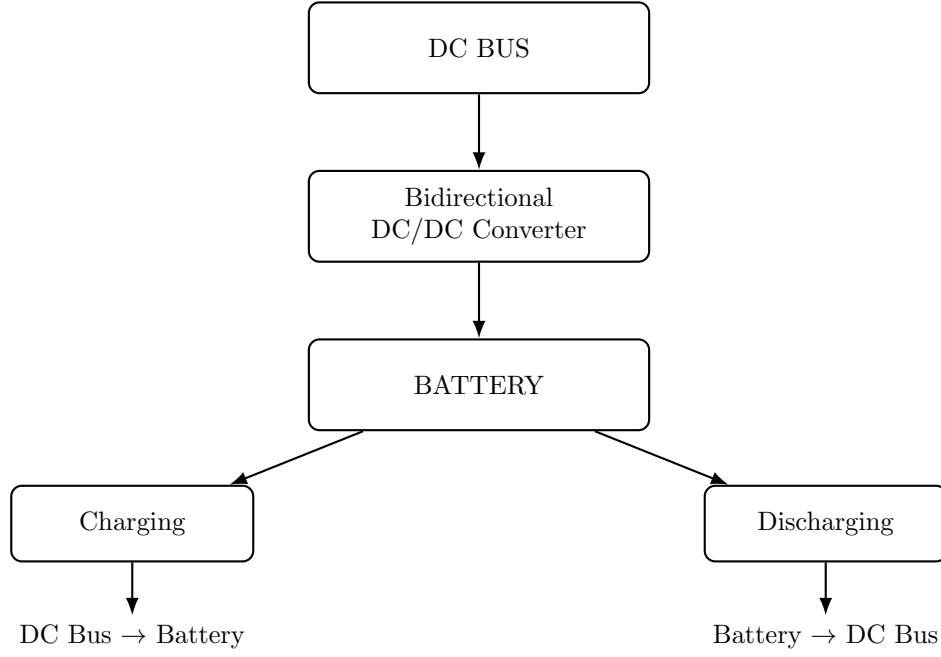


Figure 5.1: Battery energy-storage interface through the bidirectional DC/DC converter.

The bidirectional converter acts as the interface responsible for transferring electrical energy between the DC bus and the battery.

The battery therefore performs two principal functions:

5.2.1 Energy Absorption

When renewable generation exceeds the load,

$$P_G > P_L \quad (5.2)$$

the excess energy is stored in the battery.

5.2.2 Energy Supply

When renewable generation is lower than the load,

$$P_G < P_L \quad (5.3)$$

the battery supplies the required additional energy.

5.3 Battery Charging and Discharging Conditions

The basic energy-management decision is based on the difference between generated power and load power.

Define the power imbalance as

$$\Delta P = P_G - P_L \quad (5.1)$$

where:

- P_G = generated renewable power,
- P_L = load power,
- ΔP = generation-load power difference.

Three operating conditions can therefore be identified.

5.3.1 Excess Generation

If

$$\boxed{\Delta P > 0} \quad (5.4)$$

then

$$P_G > P_L. \quad (5.5)$$

The generated power is greater than the load demand.
Therefore,

$$\boxed{\text{Battery Charging}} \quad (5.6)$$

and the converter operates in the buck direction:

$$\boxed{\text{DC Bus} \rightarrow \text{Battery}} \quad (5.7)$$

5.3.2 Power Balance

If

$$\boxed{\Delta P = 0} \quad (5.8)$$

then

$$P_G = P_L. \quad (5.9)$$

The renewable system supplies the load directly and there is ideally no need for significant battery power exchange.

5.3.3 Generation Deficit

If

$$\boxed{\Delta P < 0} \quad (5.10)$$

then

$$P_G < P_L. \quad (5.11)$$

The renewable generation is insufficient to satisfy the load.
Therefore,

$$\boxed{\text{Battery Discharging}} \quad (5.12)$$

and the converter operates in the boost direction:

$$\boxed{\text{Battery} \rightarrow \text{DC Bus}} \quad (5.13)$$

The control flow starts with detection of P_G and P_L , followed by evaluation of $P_G - P_L$.

5.4 Battery State of Charge

The **State of Charge (SOC)** indicates the amount of charge stored in the battery relative to its rated capacity.

The general definition is

$$\boxed{SOC(t) = \frac{Q(t)}{Q_I}} \quad (5.2)$$

where:

- $Q(t)$ = available charge at time t ,

- Q_I = rated battery capacity.

If SOC is expressed as a percentage,

$$\boxed{SOC_{\%}(t) = 100 \frac{Q(t)}{Q_I}} \quad (5.3)$$

If SOC is expressed in per-unit form,

$$0 \leq SOC \leq 1. \quad (5.14)$$

Thus,

$$SOC = 0 \quad (5.15)$$

represents an empty battery in the idealized model, while

$$SOC = 1 \quad (5.16)$$

represents the rated full-charge condition.

5.5 Ampere-Hour Method for SOC Estimation

The reference paper uses the **Ampere-Hour method** for detecting battery SOC.

The equation is

$$\boxed{SOC = \frac{Q_0 + \sigma \int_0^t i_c dt - \int_0^t i_d dt - \int_0^t S(T_a, t) dt}{Q_I}} \quad (5.4)$$

where:

- Q_0 = initial battery charge, Ah,
- σ = charging efficiency,
- i_c = charging current, A,
- i_d = discharging current, A,
- $S(T_a, t)$ = self-discharge term,
- Q_I = battery capacity, Ah.

These definitions and the Ampere-Hour formulation are taken directly from the reference material.

5.6 Interpretation of the SOC Equation

Equation (5.4) can be understood physically as follows.

The initial battery charge is

$$Q_0. \quad (5.17)$$

During charging, the accumulated charge is

$$\sigma \int_0^t i_c dt. \quad (5.18)$$

Because charging efficiency may be less than unity in a practical battery, the charging contribution is multiplied by

$$\sigma. \quad (5.19)$$

During discharging, charge is removed from the battery:

$$\int_0^t i_d dt. \quad (5.20)$$

The self-discharge contribution is represented by

$$\int_0^t S(T_a, t) dt. \quad (5.21)$$

Therefore,

Available charge = Initial charge + Charged amount – Discharged amount – Self-discharge.

(5.22)

Dividing this quantity by the rated battery capacity gives the normalized SOC.

5.7 Simplified SOC Model Used in the Reference Simulation

The reference paper makes the following assumptions:

$$\boxed{Q_0 = 0} \quad (5.23)$$

$$\boxed{\sigma = 1} \quad (5.24)$$

$$\boxed{S(T_a, t) = 0} \quad (5.25)$$

and

$$\boxed{Q_I = 2 \text{ Ah}} \quad (5.26)$$

The charging process and battery model are assumed ideal for this simulation.

Under these assumptions, Equation (5.4) simplifies to

$$\boxed{SOC(t) = \frac{\int_0^t i_c dt - \int_0^t i_d dt}{Q_I}} \quad (5.5)$$

For

$$Q_I = 2 \text{ Ah}, \quad (5.27)$$

the SOC can therefore be obtained by integrating the net battery current and normalizing it by the battery capacity.

5.8 SOC Estimation Block

The SOC estimator can be represented as shown in Figure 5.2.

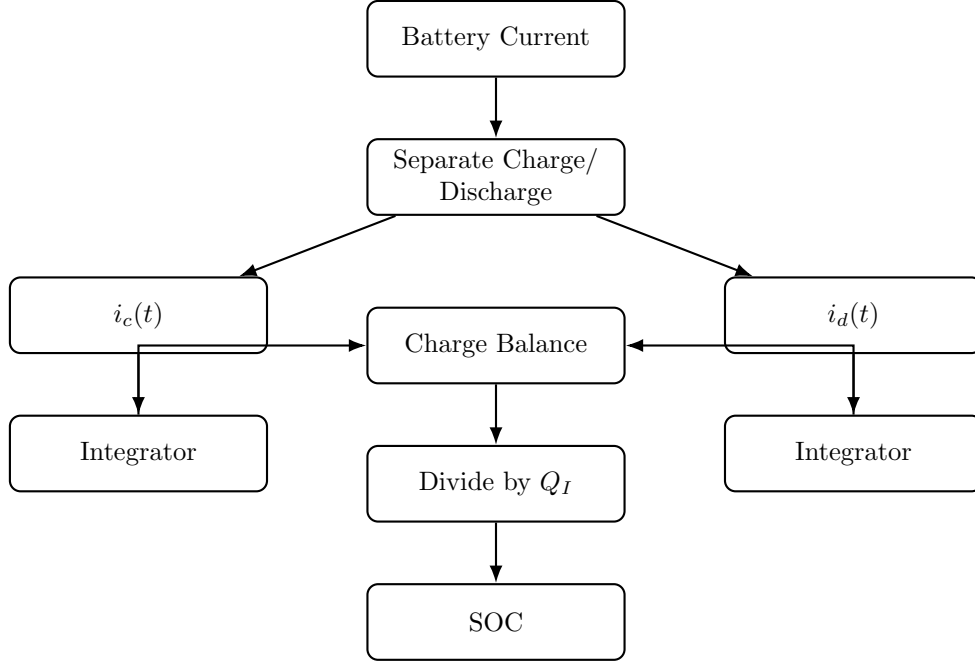


Figure 5.2: Conceptual SOC estimation block based on the Ampere-Hour method.

The reference paper implements separate SOC-detection models for charging and discharging and combines them into an SOC detection module.

5.9 SOC During Charging

During charging,

$$i_c > 0. \quad (5.28)$$

The stored charge increases with time:

$$\frac{dQ}{dt} = \sigma i_c. \quad (5.29)$$

For the idealized reference case where

$$\sigma = 1, \quad (5.30)$$

we obtain

$$\boxed{\frac{dQ}{dt} = i_c} \quad (5.6)$$

Consequently,

$$\boxed{\frac{dSOC}{dt} = \frac{i_c}{Q_I}} \quad (5.7)$$

Therefore, a positive charging current causes the SOC to increase.

5.10 SOC During Discharging

During discharging, charge is removed from the battery.

The corresponding charge variation is represented by

$$\frac{dQ}{dt} = -i_d. \quad (5.31)$$

Therefore,

$$\boxed{\frac{dSOC}{dt} = -\frac{i_d}{Q_I}} \quad (5.8)$$

Thus, increasing discharge current causes the SOC to decrease more rapidly.

This provides the mathematical connection between the battery current and SOC trajectory.

5.11 SOC-Based Energy Management

The reference work uses an SOC-based approach for charge/discharge management. It describes the charge/discharge scheduling as a **dynamic-programming method based on SOC**, with the advantage of considering battery physical constraints.

The simplified control flow is shown in Figure 5.3.

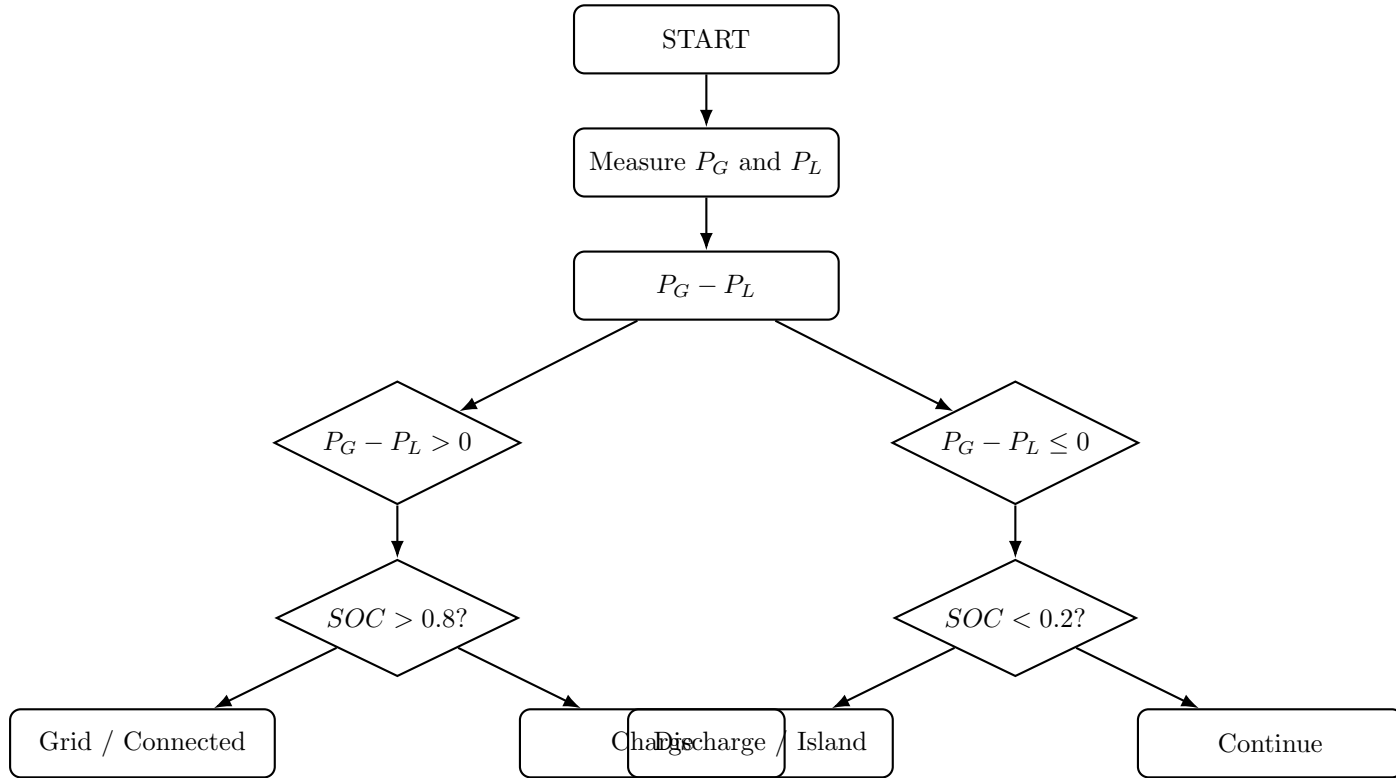


Figure 5.3: Simplified SOC-based energy-management decision process.

The extracted control-flow diagram identifies $P_G - P_L > 0$, $SOC > 0.8$, and $SOC < 0.2$ as decision conditions.

5.12 Upper SOC Limit

The upper SOC threshold used in the reference control flow is

$$\boxed{SOC_{\max} = 0.8} \quad (5.9)$$

Therefore,

$$SOC > 0.8 \quad (5.32)$$

indicates that the battery has reached the upper SOC region.

When the system is in a charging condition and the SOC reaches this limit, further charging should be stopped or prevented by the energy-management logic.

The reference simulation demonstrates this behaviour: when the battery reaches its upper charging limit, the charging current becomes zero.

5.13 Lower SOC Limit

The lower threshold is

$$\boxed{SOC_{\min} = 0.2} \quad (5.10)$$

Thus,

$$SOC < 0.2 \quad (5.33)$$

indicates that the battery has reached its lower SOC region.

Further discharge should therefore be prevented or limited.

The reference control flow explicitly includes the $SOC < 0.2$ condition in the discharge decision.

5.14 Complete Energy-Management Logic

The complete logic is summarized in Table 5.1.

Table 5.1: Complete energy-management logic.

Condition	SOC Condition	Action	Converter Mode
$P_G > P_L$	$SOC < 0.8$	Charge battery	Buck
$P_G > P_L$	$SOC \geq 0.8$	Stop/limit charging	No charging
$P_G < P_L$	$SOC > 0.2$	Discharge battery	Boost
$P_G < P_L$	$SOC \leq 0.2$	Stop/limit discharge	No discharge
$P_G = P_L$	Any permitted SOC	No major battery exchange	Idle

This table connects the **power-balance condition**, **SOC constraint**, and **converter operating mode**.

5.15 Relationship Between Energy Management and Converter Control

The battery-management system does not directly regulate the semiconductor switches. Instead, it determines the desired operating mode.

The resulting hierarchy is illustrated in Figure 5.4.

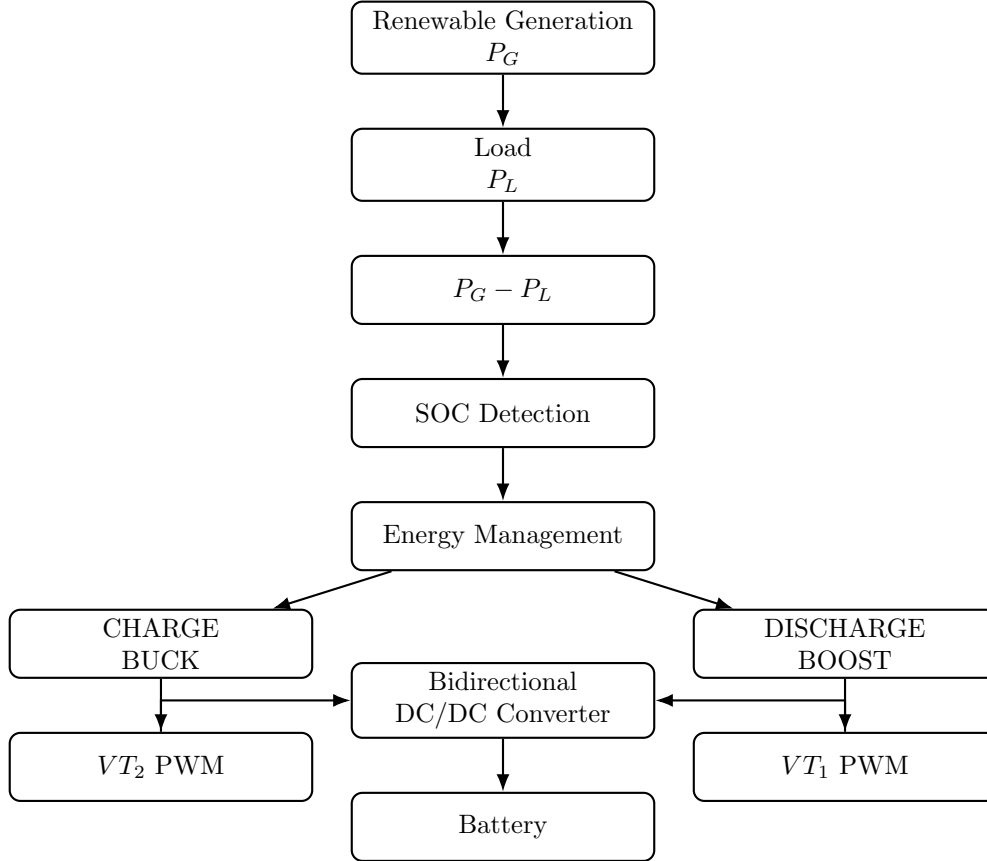


Figure 5.4: Relationship between energy management and converter control.

This diagram shows that **SOC management is the supervisory layer**, while PWM control is the lower-level converter-control layer.

5.16 Battery Power

Battery power can be expressed as

$$P_B = V_B I_B \quad (5.11)$$

where:

- V_B = battery voltage,
- I_B = battery current.

However, the sign of P_B depends on the current reference convention adopted in the Simulink model. Therefore, the sign convention must be explicitly defined in the report.

For example:

$$P_B > 0 \Rightarrow \text{Battery Discharging} \quad (5.34)$$

and

$$P_B < 0 \Rightarrow \text{Battery Charging.} \quad (5.35)$$

Important: Do not automatically use this convention if the Simulink model defines the opposite sign. Consistency between the equation, plots and discussion is required.

5.17 Battery Current and Converter Operation

The battery current provides a direct indication of the direction of energy exchange.

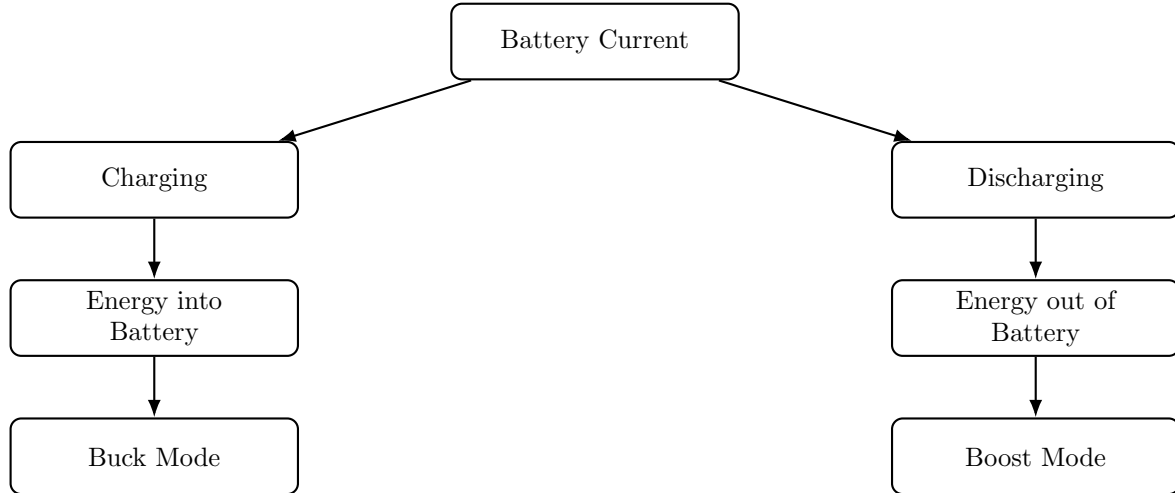


Figure 5.5: Relationship between battery current and converter operating mode.

The reference simulation shows current reversal during the transition from charging to discharging. This current reversal is an important simulation signature of bidirectional operation.

5.18 Reference Simulation Behaviour

The reference paper provides a useful example of the battery-management response.

During the charging interval, the battery begins charging when system generation exceeds the load. The battery voltage rises and stabilizes around 45 V, while charging current gradually decreases. When the upper charging limit is reached, the charging current becomes zero and the battery voltage is maintained around 40 V.

During the later deficit condition, the battery begins discharging. The battery voltage decreases and the current reverses direction. When the battery reaches the lower critical condition, discharge is stopped and the battery voltage stabilizes around 26 V.

Important: These values should be presented in the report as reference-paper results, not as your own simulation results unless your simulation reproduces them.

5.19 SOC Validation

The reference work compares the SOC obtained from its SOC detection algorithm with the SOC measurement from the Simulink battery model.

The two SOC curves have almost the same trend, with small differences attributed to the behaviour of the Simulink battery model during overcharging.

For your own simulation, produce a similar comparison:

$$\boxed{SOC_{\text{estimated}} \quad \text{vs.} \quad SOC_{\text{battery}}} \quad (5.36)$$

The SOC estimation error can be calculated as

$$\boxed{e_{SOC}(t) = SOC_{\text{battery}}(t) - SOC_{\text{estimated}}(t)} \quad (5.12)$$

Then evaluate

$$\boxed{e_{SOC,\text{max}} = \max |e_{SOC}(t)|} \quad (5.37)$$

This provides a quantitative validation rather than relying only on visual comparison.

5.20 Complete Battery-Management Block Diagram

The complete battery-management structure is shown in Figure 5.6.

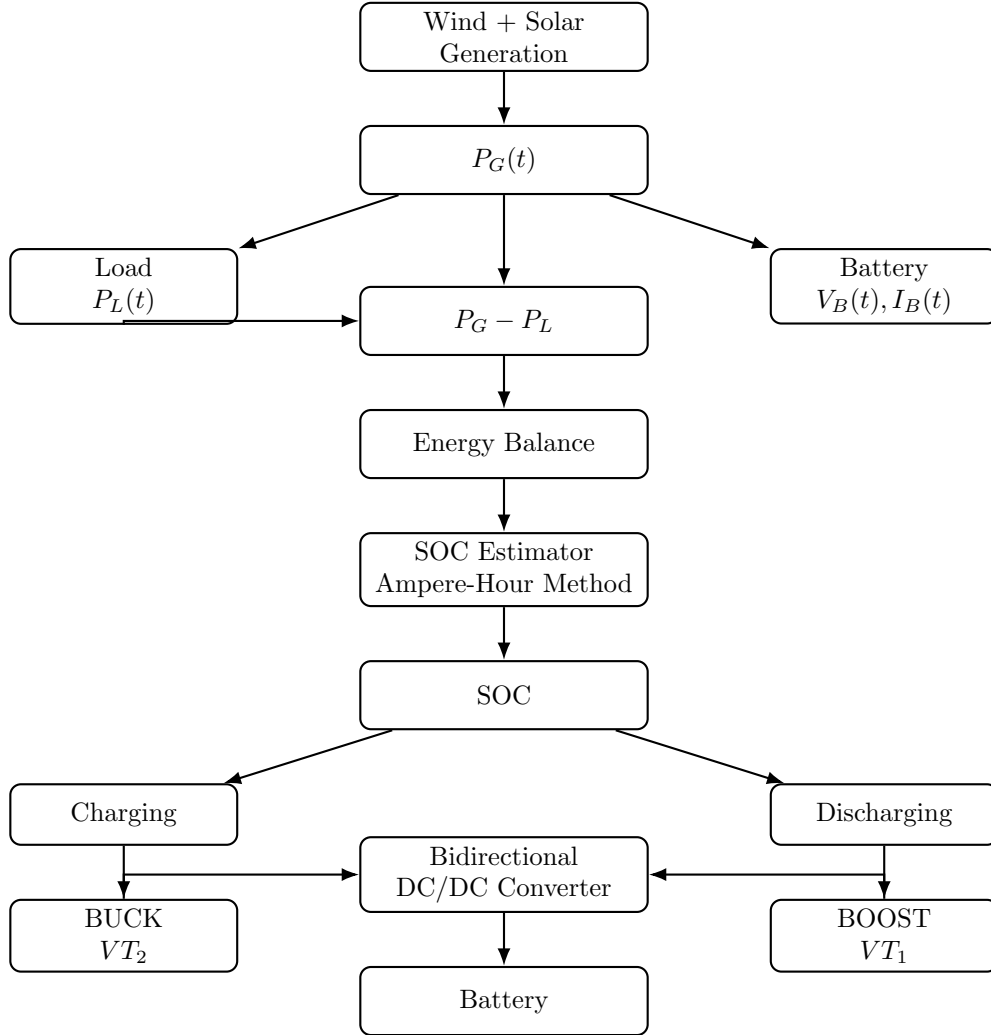


Figure 5.6: Complete battery-management and SOC-control structure.

5.21 Chapter 5 Notations

The principal symbols used in this chapter are summarized in Table 5.2.

Table 5.2: Notations used in Chapter 5.

Symbol	Description	Unit
SOC	State of charge	p.u. / %
$Q(t)$	Available battery charge	Ah
Q_0	Initial battery charge	Ah
Q_I	Rated battery capacity	Ah
i_c	Charging current	A
i_d	Discharging current	A
σ	Charging efficiency	–
$S(T_a, t)$	Self-discharge term	Ah
V_B	Battery voltage	V
I_B	Battery current	A
P_B	Battery power	W
P_G	Generated power	W
P_L	Load power	W
ΔP	Generation-load power difference	W
$SOC_{\max}$	Upper SOC threshold	p.u.
$SOC_{\min}$	Lower SOC threshold	p.u.
e_{SOC}	SOC estimation error	p.u. / %

Final Root Map for Chapter 5

The complete chapter can be summarized by the following root map.

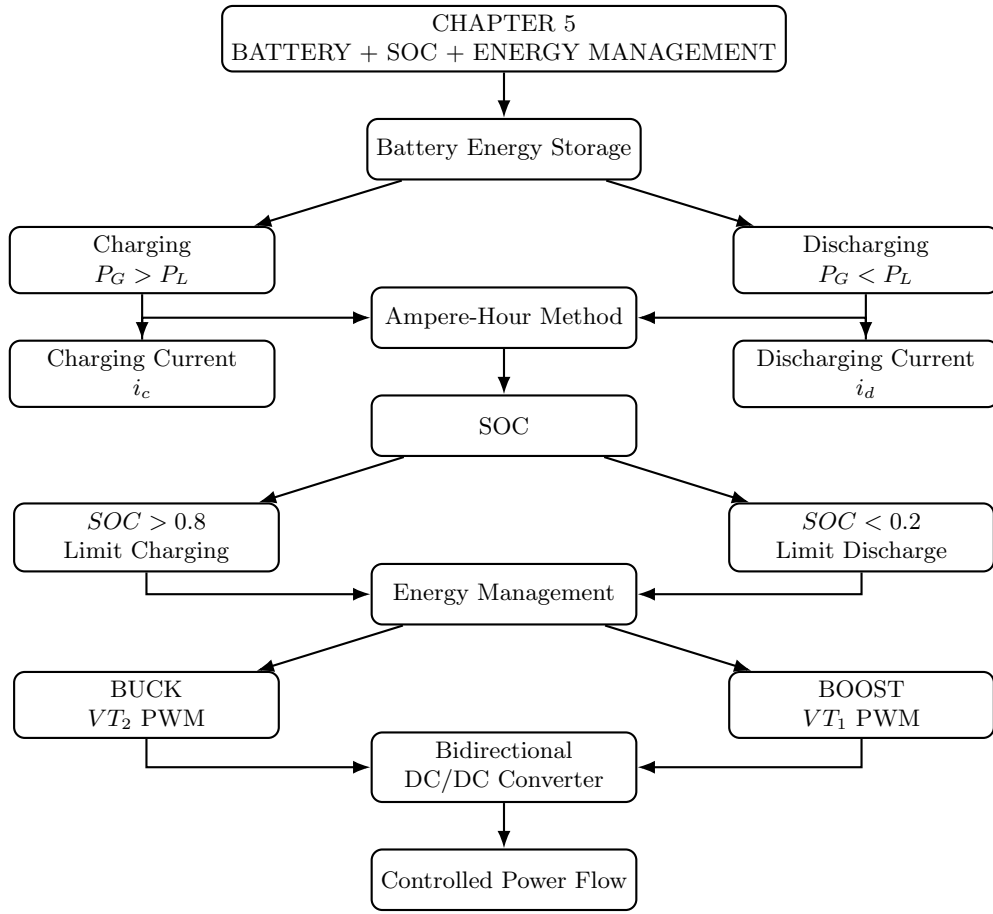


Figure 5.7: Root map of Chapter 5: Battery, SOC estimation and energy management.

Important Distinction in the Control Architecture

The reference material describes two related but distinct control levels.

First, the energy-management layer determines the charging or discharging decision:

$$\boxed{\text{Energy Management} \rightarrow \text{SOC-based Decision} / \text{Dynamic Programming}} \quad (5.38)$$

Second, the converter-control layer implements the selected operating mode:

$$\boxed{\text{Converter Control} \rightarrow \text{Adaptive} / \text{PI-PID-based Duty Control}} \quad (5.39)$$

Therefore, the SOC-management algorithm should not be confused with the low-level converter controller.

Chapter 6

MATLAB/Simulink Model Development and Simulation

6.1 Introduction

This chapter presents the development of the complete MATLAB/Simulink model of the bidirectional DC/DC converter and its integration with the battery energy-storage system.

The primary objective of the simulation is to verify the practical operation of the converter and the associated energy-management system under different renewable-generation and load conditions.

The simulation is developed to verify the following major functionalities:

- bidirectional energy transfer between the DC bus and the battery;
- battery charging when renewable generation exceeds the load demand;
- battery discharging when renewable generation is insufficient to satisfy the load;
- battery state-of-charge (SOC) estimation;
- DC-bus voltage regulation around its desired reference value;
- automatic transition between buck and boost operating modes;
- appropriate PWM generation for the semiconductor switches; and
- prevention of simultaneous conduction of the two semiconductor devices.

The reference work implements the bidirectional converter and its adaptive-control algorithm using MATLAB/Simulink and uses simulation to verify the charging and discharging operation of the energy-storage system.

Therefore, this chapter establishes the connection between the mathematical models developed in the preceding chapters and the simulation results obtained from the complete system.

The overall simulation philosophy can be expressed as

Mathematical Model $\rightarrow$ Control Strategy $\rightarrow$ MATLAB/Simulink Model $\rightarrow$ Simulation $\rightarrow$ Performance Evaluation

(6.1)

6.2 MATLAB/Simulink Simulation Environment

The complete simulation consists of several interconnected subsystems. The principal components are the renewable-generation system, DC bus, electrical load, battery, bidirectional DC/DC converter and controller.

The complete simulation structure can be represented as

Renewable Generation + DC Bus + Load + Battery + Bidirectional Converter + Controller

(6.2)

The MATLAB/Simulink implementation can be organized into the following functional stages:

1. renewable-generation subsystem;
2. DC-bus subsystem;
3. load subsystem;
4. bidirectional DC/DC converter;
5. power-balance detection;
6. SOC estimation;
7. energy-management subsystem;
8. buck controller;
9. boost controller;
10. PWM generation;
11. switching-state protection;
12. battery model; and
13. measurement and feedback system.

The high-level control and power-flow relationship is

$$\boxed{P_G - P_L \rightarrow SOC \rightarrow \text{Energy Management} \rightarrow \text{Buck/Boost Selection} \rightarrow \text{PWM} \rightarrow VT_1, VT_2} \quad (6.3)$$

where P_G represents the generated renewable power and P_L represents the load power.

The simulation is therefore not treated as a single monolithic model. Instead, it is divided into subsystems so that the electrical power stage and control system can be tested independently before the final system-level simulation.

6.3 Simulation Model of the Energy-Storage Module

The reference work develops the simulation model according to the main circuit configuration of the energy-storage module.

The model contains the principal elements required to reproduce the energy-storage behaviour:

1. battery/source model;
2. bidirectional DC/DC converter;
3. converter inductor;
4. converter capacitors;
5. semiconductor switching devices;
6. electrical load;
7. control system;
8. voltage and current measurement blocks;
9. SOC estimation subsystem; and
10. feedback paths.

The simulation model is preferably constructed hierarchically. Separate subsystems are used for the converter, controller, battery, SOC estimation and energy-management system.

This hierarchical implementation has several advantages:

- individual subsystems can be tested independently;
- errors can be isolated more easily;
- the complete model becomes easier to understand;
- controller modifications can be implemented without rebuilding the complete electrical circuit; and
- the model can be explained more clearly during technical presentations and viva examinations.

6.4 Simulink Model Hierarchy

For an organized MATLAB/Simulink implementation, the complete model can be divided into several hierarchical subsystems.

The main power-flow structure is

$$\boxed{\text{Renewable Sources} \rightarrow \text{DC Bus} \rightarrow \text{Load}} \quad (6.4)$$

with the battery connected to the DC bus through the bidirectional converter:

$$\boxed{\text{DC Bus} \leftrightarrow \text{Bidirectional DC/DC Converter} \leftrightarrow \text{Battery}} \quad (6.5)$$

The control subsystem operates in parallel with the electrical power stage. The complete hierarchy can therefore be described as

$$\boxed{\begin{array}{l} \text{Main Model} \rightarrow \text{Power System} \\ \quad + \text{ Converter} \\ \quad + \text{ Control} \\ \quad + \text{ Battery} \\ \quad + \text{ SOC} \\ \quad + \text{ Measurement.} \end{array}} \quad (6.6)$$

The control path follows

$$P_G, P_L \rightarrow P_G - P_L \rightarrow \text{SOC} \rightarrow \text{Mode Selection} \rightarrow \text{Buck/Boost Controller.} \quad (6.7)$$

This arrangement allows the supervisory energy-management system to determine the required direction of power flow, while the converter controller generates the required duty ratio.

6.5 Bidirectional DC/DC Converter Subsystem

The bidirectional DC/DC converter is the central power-electronic interface between the DC bus and the battery.

The principal semiconductor switches are

$$VT_1 \quad (6.8)$$

and

$$VT_2. \quad (6.9)$$

The converter supports two fundamental operating modes:

1. buck or step-down operation; and
2. boost or step-up operation.

6.5.1 Buck Operating Mode

When the battery is being charged, the converter operates in step-down or buck mode.

The corresponding switching condition is

$$\boxed{VT_1 = OFF, \quad VT_2 = ON/PWM.} \quad (6.10)$$

The direction of power flow is

$$\boxed{\text{DC Bus} \rightarrow \text{Battery.}} \quad (6.11)$$

This operating condition occurs when

$$\boxed{P_G > P_L.} \quad (6.12)$$

Therefore, the excess renewable energy is transferred to the battery.

6.5.2 Boost Operating Mode

When the battery supplies energy to the DC bus, the converter operates in step-up or boost mode.

The corresponding switching condition is

$$\boxed{VT_1 = ON/PWM, \quad VT_2 = OFF.} \quad (6.13)$$

The direction of power flow becomes

$$\boxed{\text{Battery} \rightarrow \text{DC Bus.}} \quad (6.14)$$

This operating condition occurs when

$$\boxed{P_G < P_L.} \quad (6.15)$$

The battery therefore compensates for the power deficit.

6.6 Switching-State Protection

Safe switching of the semiconductor devices is an essential requirement of the bidirectional converter.

Considering the two switches VT_1 and VT_2 , four theoretical switching combinations are possible.

Table 6.1: Switching states of the bidirectional converter.

VT_1	VT_2	Operating Condition
OFF	ON	Step-down / Buck
ON	OFF	Step-up / Boost
OFF	OFF	Inactive state
ON	ON	Forbidden state

The condition

$$VT_1 = ON, \quad VT_2 = ON \quad (6.16)$$

must be avoided.

Simultaneous conduction can create an undesirable electrical path and can cause improper operation of the converter.

Therefore, an additional switching-logic or interlock mechanism is introduced into the controller pulse-output path.

The switching logic ensures that only the appropriate switch is activated for the selected operating mode.

The fundamental condition is

$$\boxed{VT_1 \cdot VT_2 = 0} \quad (6.17)$$

for the normal operating states.

Thus,

$$VT_1 = 1 \Rightarrow VT_2 = 0 \quad (6.18)$$

and

$$VT_2 = 1 \Rightarrow VT_1 = 0. \quad (6.19)$$

This interlock is particularly important during the transition between buck and boost modes.

6.7 Inductor Design Used in the Simulation

The converter inductor is an important energy-storage element in both operating modes.

It performs two primary functions:

- storing energy during part of the switching cycle; and
- transferring energy between the DC bus and battery.

The inductance also influences the current ripple and dynamic behaviour of the converter.

The reference work obtains different inductance values for the buck and boost operating conditions.

For buck operation,

$$L_{\text{buck}} = 350 \times 10^{-6} \text{ H} \quad (6.20)$$

or

$$L_{\text{buck}} = 350 \text{ } \mu\text{H}. \quad (6.21)$$

For boost operation,

$$L_{\text{boost}} = 104 \times 10^{-6} \text{ H} \quad (6.22)$$

or

$$L_{\text{boost}} = 104 \text{ } \mu\text{H}. \quad (6.23)$$

These values correspond to the reference-paper design. If the present MATLAB/Simulink implementation uses different inductance values, those values should be reported separately as the actual simulation parameters.

The two values should therefore not be confused with a universal inductance requirement for all bidirectional converter designs.

6.8 PWM Implementation in Simulink

The controller output is converted into switching pulses using a pulse-width modulation subsystem.

The PWM process can be represented as

$$\text{Controller Output} \rightarrow \text{Carrier Comparison} \rightarrow \text{PWM Signal} \rightarrow \text{Switching Logic} \rightarrow VT_1, VT_2. \quad (6.24)$$

The reference implementation uses a triangular carrier with amplitude

$$A_c = 1 \quad (6.25)$$

and switching frequency

$$f_{sw} = 20 \text{ kHz}. \quad (6.26)$$

The corresponding switching period is

$$T_{sw} = \frac{1}{f_{sw}} \quad (6.27)$$

and therefore

$$T_{sw} = \frac{1}{20 \times 10^3}. \quad (6.28)$$

Hence,

$$T_{sw} = 50 \text{ } \mu\text{s}. \quad (6.1)$$

The modulating signal generated by the controller is compared with the carrier signal to generate the required PWM pulse.

The PWM signal is then passed through the switching-protection logic before being applied to the semiconductor devices.

6.9 Buck Controller Simulation

The buck controller is activated when the battery is required to absorb excess renewable energy.

The basic control sequence is

$$\boxed{V_o^* \rightarrow e(t) \rightarrow \text{PI/PID} \rightarrow D \rightarrow \text{PWM} \rightarrow VT_2.} \quad (6.29)$$

Here, V_o^* represents the desired output voltage and $e(t)$ represents the voltage error.

The reference implementation defines the voltage error as

$$\boxed{e(t) = U_o(t) - U'_o(t).} \quad (6.30)$$

The controller processes this error and generates the appropriate duty ratio.

The resulting modulation signal is compared with the triangular carrier to generate the PWM signal.

During buck operation,

$$VT_1 = OFF \quad (6.31)$$

while

$$VT_2 = ON/PWM. \quad (6.32)$$

The converter therefore transfers energy from the DC bus toward the battery.

The buck controller must maintain stable battery-side operation while limiting current and voltage variations.

6.10 Boost Controller Simulation

The boost controller is activated when the battery must supply energy to the DC bus.

The basic control sequence is

$$\boxed{V_{DC} \rightarrow e_D \rightarrow \text{PI/PID} \rightarrow D \rightarrow \text{PWM} \rightarrow VT_1.} \quad (6.33)$$

The desired duty ratio for an ideal boost converter is given by

$$\boxed{D = 1 - \frac{U_i}{U_o}.} \quad (6.34)$$

The actual duty ratio can be expressed as

$$\boxed{D' = 1 - \frac{U_i}{U'_o}.} \quad (6.35)$$

The duty-ratio error is consequently related to the difference between the desired and actual converter operating conditions.

The controller modifies the duty ratio so that the output voltage moves toward its reference.

The boost duty ratio is limited according to

$$\boxed{0 \leq D \leq 0.9.} \quad (6.36)$$

The duty-ratio limit prevents the controller from operating in an undesirable region and helps maintain an acceptable dynamic response.

During boost operation,

$$VT_1 = ON/PWM \quad (6.37)$$

and

$$VT_2 = OFF. \quad (6.38)$$

Consequently,

$$\boxed{\text{Battery} \rightarrow \text{DC Bus.}} \quad (6.39)$$

6.11 Battery State-of-Charge Simulation

The battery SOC is an important supervisory variable in the energy-management system.

The SOC subsystem is developed separately from the low-level converter controller.

The battery current is processed according to the charging or discharging condition and integrated to determine the accumulated charge.

The general SOC relationship can be expressed as

$$SOC(t) = SOC(0) + \frac{1}{Q_I} \int_0^t i_{\text{net}}(\tau) d\tau \quad (6.40)$$

where Q_I represents the nominal battery capacity and i_{net} represents the appropriately signed battery current.

For the charging and discharging processes, the sign convention must be applied consistently.

During charging,

$$SOC \uparrow \quad (6.41)$$

whereas during discharging,

$$SOC \downarrow. \quad (6.42)$$

The reference implementation uses separate SOC-estimation branches for charging and discharging.

6.12 SOC Detection Structure

The SOC subsystem can be represented conceptually as

$$I_B \rightarrow \begin{cases} \text{Charging Branch,} \\ \text{Discharging Branch} \end{cases} \rightarrow \text{Integration} \rightarrow \text{Charge Balance} \rightarrow SOC. \quad (6.43)$$

The charging branch integrates the charging current contribution, while the discharging branch accounts for the charge removed from the battery.

The two contributions are then combined to obtain the battery SOC.

The estimated SOC is subsequently provided to the energy-management system.

Thus,

$$SOC \rightarrow \text{Energy Management} \rightarrow \text{Charge/Discharge Decision}. \quad (6.44)$$

6.13 SOC Comparison and Validation

An important part of the simulation is the validation of the SOC estimation subsystem.

The estimated SOC is compared with the SOC obtained from the battery model.

The two quantities are represented as

$$SOC_{\text{estimated}} \quad (6.45)$$

and

$$SOC_{\text{actual}}. \quad (6.46)$$

The SOC estimation error is defined as

$$e_{SOC}(t) = SOC_{\text{actual}}(t) - SOC_{\text{estimated}}(t). \quad (6.4)$$

The maximum absolute SOC error is

$$e_{SOC,\text{max}} = \max_t |e_{SOC}(t)|. \quad (6.5)$$

The reference simulation indicates that the estimated SOC and battery-model SOC follow nearly the same overall trend.

Small differences can occur because of the behaviour of the Simulink battery model, particularly around the charging and overcharge regions.

Therefore, SOC validation should consider both:

1. the visual agreement between the two SOC trajectories; and
2. the quantitative SOC estimation error.

6.14 DC-Bus Voltage Regulation

The DC bus forms the common electrical interface between the renewable-generation system, load and battery energy-storage system.

The reference DC-bus voltage is

$$\boxed{V_{DC}^* = 300 \text{ V.}} \quad (6.47)$$

Maintaining the DC-bus voltage near this value is important because large voltage variations can affect the energy-flow analysis and the operation of the connected subsystems.

The reference system therefore introduces an additional voltage regulator between the generation unit and the DC bus.

The overall relationship is

$$\boxed{\text{Wind/Solar Generation} \rightarrow \text{Voltage Regulator} \rightarrow \text{DC Bus}} \quad (6.48)$$

with the load and battery connected to the DC bus.

The DC-bus voltage error can be expressed as

$$\boxed{e_V(t) = V_{DC}^* - V_{DC}(t).} \quad (6.2)$$

The controller acts on this error to reduce the deviation of the DC-bus voltage from its reference.

6.15 DC-Bus Voltage Regulation Result

The reference simulation demonstrates that the DC-bus voltage becomes stable after the generator terminal voltage reaches an appropriate operating level.

The desired DC-bus voltage is

$$V_{DC}^* = 300 \text{ V.} \quad (6.49)$$

The reference result indicates that the DC-bus voltage fluctuates around

$$\boxed{V_{DC} \approx 300 \text{ V.}} \quad (6.50)$$

The reference paper also reports that the regulator can restore the output voltage to the desired 300-V value within approximately

$$\boxed{t_r \approx 10 \text{ ms}} \quad (6.51)$$

following a step input.

This recovery time is an important indicator of the dynamic performance of the voltage-regulation subsystem.

6.16 Complete System Co-Simulation

After independently validating the energy-storage subsystem, it is integrated with the complete wind-solar hybrid energy system.

The complete simulation contains

$$\boxed{\text{Wind} + \text{Solar} + \text{DC Bus} + \text{Load} + \text{Battery} + \text{Bidirectional DC/DC Converter.}} \quad (6.52)$$

The renewable sources supply power to the DC bus, while the battery acts as an energy buffer.

The battery can either absorb excess power or supply power to compensate for a generation deficit.

The supervisory control therefore follows

$$P_G - P_L \rightarrow \text{Energy Management} \rightarrow \text{Converter Mode} \rightarrow \text{Battery Power.} \quad (6.53)$$

The complete system must coordinate the following variables:

- renewable-generation power P_G ;
- load power P_L ;
- battery voltage V_B ;
- battery current I_B ;
- battery power P_B ;
- battery SOC;
- DC-bus voltage V_{DC} ;
- converter duty ratio D ; and
- semiconductor switching states.

6.17 Power-Balance Simulation

The fundamental power-balance variable is defined as

$$\Delta P = P_G - P_L. \quad (6.6)$$

The sign of ΔP determines the required energy-storage operation.

6.17.1 Case 1: Excess Renewable Generation

When

$$P_G > P_L, \quad (6.54)$$

the renewable system produces more power than required by the load.

The excess power is directed toward the battery.

Therefore,

$$P_G > P_L \Rightarrow \text{Battery Charging.} \quad (6.55)$$

The bidirectional converter operates in buck mode.

The battery SOC increases:

$$SOC \uparrow. \quad (6.56)$$

6.17.2 Case 2: Generation Deficit

When

$$P_G < P_L, \quad (6.57)$$

the renewable generation cannot completely satisfy the load.

The battery supplies the remaining power.

Thus,

$$P_G < P_L \Rightarrow \text{Battery Discharging.} \quad (6.58)$$

The bidirectional converter operates in boost mode.

The battery SOC consequently decreases:

$$SOC \downarrow. \quad (6.59)$$

6.17.3 Case 3: Approximate Power Balance

When

$$\boxed{P_G \approx P_L,} \quad (6.60)$$

the renewable system approximately satisfies the load demand.

Therefore, the battery does not need to exchange significant power with the DC bus.

This condition can be considered an approximately balanced or idle battery operating condition.

6.18 Simulation Sequence

A systematic simulation sequence is used to validate the complete system.

The simulation is divided into five major stages.

6.18.1 Stage 1: Converter Validation

The bidirectional converter is first tested independently.

The following quantities are verified:

- buck operation;
- boost operation;
- inductor current;
- battery voltage;
- battery current; and
- PWM switching signals.

The purpose of this stage is to confirm correct electrical operation before introducing the complete energy-management system.

6.18.2 Stage 2: SOC Validation

The battery model and SOC estimator are then simulated.

The primary validation condition is

$$\boxed{SOC_{\text{estimated}} \approx SOC_{\text{actual}}.} \quad (6.61)$$

The SOC error defined in Equation (6.4) is evaluated to determine the accuracy of the estimation subsystem.

6.18.3 Stage 3: DC-Bus Regulation

The DC-bus regulator is tested using a step or varying generation-side input.

The primary objective is

$$\boxed{V_{DC} \rightarrow 300 \text{ V}.} \quad (6.62)$$

The maximum voltage error and settling/recovery time are evaluated.

6.18.4 Stage 4: Integrated System

The following subsystems are then connected:

- renewable generation;
- DC bus;
- electrical load;
- battery;

- bidirectional DC/DC converter;
- SOC controller;
- energy-management system; and
- DC-bus regulator.

The purpose is to evaluate the complete energy-storage operation.

6.18.5 Stage 5: Power-Flow Transitions

The final test introduces variations in generation and load so that the battery passes through different operating states.

The desired sequence is

$$\boxed{\text{Charging} \rightarrow \text{Idle} \rightarrow \text{Discharging} \rightarrow \text{Charging.}} \quad (6.63)$$

This sequence is particularly important because it demonstrates the bidirectional nature of the converter.

6.19 Reference Simulation Timeline

The reference simulation provides several important operating events that can be used to interpret the battery voltage, current and SOC responses.

6.19.1 Initial Charging Condition

At approximately

$$t = 0.06 \text{ s}, \quad (6.64)$$

the generated power exceeds the load:

$$P_G > P_L. \quad (6.65)$$

The battery consequently begins charging.

The converter operates in buck mode and transfers energy from the DC bus toward the battery.

6.19.2 Upper Charging Condition

At approximately

$$t = 0.76 \text{ s}, \quad (6.66)$$

the battery reaches the reported upper charging limit.

The charging current becomes zero and further charging is stopped.

The reference result indicates that the battery voltage is then maintained around its corresponding operating level.

6.19.3 Change in Discharge Power Condition

The reference discussion identifies approximately

$$t = 1.35 \text{ s} \quad (6.67)$$

as a point at which the generation/load relationship changes significantly.

The discharge current becomes more pronounced when the generated power becomes considerably lower than the load.

6.19.4 Lower Battery Operating Condition

The reference paper reports that at approximately

$$t = 1.60 \text{ s}, \quad (6.68)$$

the battery reaches the lower critical operating condition.

Discharging is then stopped and the battery voltage stabilizes around

$$\boxed{V_B \approx 26 \text{ V}.} \quad (6.69)$$

6.19.5 Return Toward Charging

At approximately

$$t = 1.82 \text{ s}, \quad (6.70)$$

the reference discussion indicates a change in the power condition that allows the battery to begin charging again.

The battery current consequently changes direction toward the charging condition.

6.19.6 Timing Consistency Note

The reference text contains an ordering inconsistency because it mentions a generation/load transition at approximately 1.96 s while also describing the lower critical condition at approximately 1.60 s.

Therefore, these reported values should be treated as reference timings rather than forcing a strictly chronological interpretation.

For the final report, the exact transition times should preferably be obtained from the reproduced MATLAB/Simulink waveform if the present simulation is available.

6.20 Simulation Performance Evaluation

The simulation analysis should not be limited to visual inspection of the waveforms.

Quantitative performance measures should also be calculated.

The principal performance measures considered in this chapter are:

- DC-bus voltage error;
- maximum DC-bus voltage deviation;
- SOC estimation error;
- maximum SOC estimation error; and
- DC-bus settling/recovery time.

6.20.1 DC-Bus Voltage Error

The DC-bus voltage error is defined as

$$\boxed{e_V(t) = V_{DC}^* - V_{DC}(t).} \quad (6.2)$$

The maximum absolute voltage error is

$$\boxed{e_{V,\max} = \max_t |e_V(t)|.} \quad (6.3)$$

A smaller value of $e_{V,\max}$ indicates closer regulation of the DC bus to its desired reference.

6.20.2 SOC Estimation Error

The SOC estimation error is

$$e_{SOC}(t) = SOC_{\text{actual}}(t) - SOC_{\text{estimated}}(t). \quad (6.4)$$

The maximum absolute SOC error is

$$e_{SOC,\text{max}} = \max_t |e_{SOC}(t)|. \quad (6.5)$$

This quantity provides a numerical measure of the agreement between the estimated and actual SOC trajectories.

6.20.3 DC-Bus Settling Time

The DC-bus settling time can be defined as the time required for the voltage to enter and remain inside a specified tolerance band around its reference value.

For a $\pm 2\%$ tolerance band around 300 V,

$$300(1 \pm 0.02) \quad (6.71)$$

gives

$$294 \text{ V} \leq V_{DC} \leq 306 \text{ V}. \quad (6.72)$$

The settling time is the time required for the DC-bus voltage to enter this band and remain within it. This definition should be used consistently when evaluating the present simulation.

6.21 Expected Physical Behaviour

A successful simulation should reproduce the physical sequence associated with the bidirectional energy-storage operation.

When

$$P_G > P_L, \quad (6.73)$$

excess renewable power is available.

Therefore,

$$P_G > P_L \rightarrow \text{Excess Power} \rightarrow \text{Buck Mode} \rightarrow \text{Battery Charging}. \quad (6.74)$$

During charging,

$$SOC \uparrow. \quad (6.75)$$

When the battery reaches the upper SOC limit,

$$SOC \rightarrow SOC_{\text{max}}, \quad (6.76)$$

charging is stopped or restricted.

When the generated power becomes lower than the load,

$$P_G < P_L, \quad (6.77)$$

a power deficit occurs.

Consequently,

$$P_G < P_L \rightarrow \text{Power Deficit} \rightarrow \text{Boost Mode} \rightarrow \text{Battery Discharging}. \quad (6.78)$$

During discharging,

$$SOC \downarrow. \quad (6.79)$$

When the lower SOC limit is reached, further discharge is restricted.

Therefore, the complete physical sequence is

Generation > Load → Buck → Charging → $SOC \uparrow$ → $SOC_{\max}$ → Charging Stops → Generation < Load → Boost → Discharging → $SOC \downarrow$ → $SOC_{\min}$ → Discharging Stops.	(6.80)
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6.22 Reference Results and Present Simulation

A clear distinction must be maintained between the values reported by the reference paper and the results obtained from the present MATLAB/Simulink implementation.

The reference paper reports approximately

$$\boxed{V_{DC} \approx 300 \text{ V}} \quad (6.81)$$

with approximately

$$\boxed{10 \text{ ms}} \quad (6.82)$$

recovery to the desired DC-bus voltage after a step input.

The reference battery results include approximately

$$\boxed{V_B \approx 45 \text{ V}} \quad (6.83)$$

during charging and approximately

$$\boxed{V_B \approx 26 \text{ V}} \quad (6.84)$$

at the reported lower operating condition.

The reference simulation also demonstrates:

- decreasing charging current as the battery approaches its upper limit;
- termination of charging at the upper condition;
- reversal of current during transition to discharging;
- increasing discharge current during a stronger power deficit; and
- termination of discharge at the lower operating condition.

These values must be identified as **reference results** unless they are independently obtained from the present MATLAB/Simulink model.

The recommended comparison structure is

Reference Result	$\leftrightarrow$	Present Simulation	$\rightarrow$	Difference Analysis.	(6.85)
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6.23 Reference Parameter and Simulation Result Comparison

The following table can be used to distinguish the reference parameters from the actual results obtained from the present simulation.

Table 6.2: Reference parameters and present simulation quantities.

Parameter	Reference/Target	Present Simulation
DC-bus voltage	300 V	To be obtained
Switching frequency	20 kHz	To be obtained
PWM carrier amplitude	$A_c = 1$	To be obtained
Buck inductance	350 μH	To be obtained
Boost inductance	104 μH	To be obtained
Maximum boost duty ratio	0.9	To be obtained
Upper SOC limit	0.8	To be obtained
Lower SOC limit	0.2	To be obtained
Battery charging voltage	≈ 45 V	To be obtained
Lower battery voltage	≈ 26 V	To be obtained
DC-bus recovery time	≈ 10 ms	To be calculated
Maximum voltage error	–	To be calculated
Maximum SOC error	–	To be calculated

This format prevents reference-paper values from being mistaken for results obtained from the present simulation.

6.24 Discussion

The MATLAB/Simulink implementation demonstrates the interaction between the renewable-generation system, DC bus, battery, bidirectional converter, SOC estimator, energy-management system and DC-bus regulator.

The converter changes its operating direction according to the power imbalance between renewable generation and load.

When

$$P_G > P_L, \quad (6.86)$$

the battery absorbs excess renewable energy through buck operation.

Conversely, when

$$P_G < P_L, \quad (6.87)$$

the battery supplies the power deficit through boost operation.

The SOC subsystem provides supervisory information for restricting excessive charging and discharging.

The DC-bus voltage regulator maintains the common DC bus around its 300-V target.

The interaction between the different subsystems can therefore be summarized as

$$\begin{aligned}
P_G, P_L &\rightarrow \text{Power Balance} \\
&\rightarrow \text{SOC} \\
&\rightarrow \text{Energy Management} \\
&\rightarrow \text{Buck/Boost} \\
&\rightarrow \text{PWM} \\
&\rightarrow \text{Bidirectional Converter} \\
&\rightarrow \text{Battery.}
\end{aligned}$$

(6.88)

The reference simulation indicates that the estimated SOC follows the battery-model SOC with only small differences.

At the system level, the reference DC-bus voltage remains around 300 V and can be restored to the desired value within approximately 10 ms following a step input.

The battery voltage and current responses demonstrate the transition between charging and discharging operation.

Consequently, the simulation framework provides a complete environment for evaluating the bidirectional DC/DC converter as an energy-storage interface.

6.25 Chapter 6 Notations

The principal notations used in this chapter are summarized in Table 6.3.

Table 6.3: Notations used in Chapter 6.

Symbol	Meaning	Unit
V_{DC}	DC-bus voltage	V
V_{DC}^*	DC-bus voltage reference	V
V_B	Battery voltage	V
I_B	Battery current	A
P_G	Renewable generation power	W
P_L	Load power	W
P_B	Battery power	W
ΔP	Generation-load power imbalance	W
D	Converter duty ratio	–
D'	Actual duty ratio	–
f_{sw}	Switching frequency	Hz
T_{sw}	Switching period	s
A_c	PWM carrier amplitude	–
L	Converter inductance	H
L_{buck}	Buck-mode inductance	H
L_{boost}	Boost-mode inductance	H
C_1	Converter capacitance 1	F
C_2	Converter capacitance 2	F
VT_1	Main semiconductor switch 1	–
VT_2	Main semiconductor switch 2	–
SOC	Battery state of charge	p.u./%
SOC_{actual}	Battery-model SOC	p.u./%
$SOC_{estimated}$	Estimated SOC	p.u./%
e_V	DC-bus voltage error	V
$e_{V,max}$	Maximum absolute DC-bus voltage error	V
e_{SOC}	SOC estimation error	p.u./%
$e_{SOC,max}$	Maximum absolute SOC estimation error	p.u./%
Q_I	Battery nominal capacity	Ah
U_i	Converter input voltage	V
U_o	Desired/output voltage	V
U'_o	Actual output voltage	V

6.26 Chapter Summary

This chapter developed the MATLAB/Simulink implementation of the bidirectional DC/DC converter and the associated battery energy-storage system.

The simulation model incorporates the bidirectional converter, PWM generation, switching-state protection, battery model, SOC estimation, energy-management system and DC-bus voltage regulation.

The converter operates in buck mode when renewable generation exceeds the load and in boost mode when the battery is required to supply the load-side power deficit.

The reference implementation uses a triangular PWM carrier with

$$A_c = 1 \quad (6.89)$$

and a switching frequency of

$$f_{sw} = 20 \text{ kHz}. \quad (6.90)$$

The corresponding switching period is

$$T_{sw} = 50 \text{ } \mu\text{s}. \quad (6.91)$$

The boost duty ratio is restricted to

$$0 \leq D \leq 0.9. \quad (6.92)$$

Separate SOC-estimation branches are used for charging and discharging, and the estimated SOC is compared with the battery model SOC for validation.

The complete system is operated with a 300-V DC bus. An additional voltage-regulation subsystem is used to maintain the DC-bus voltage near its reference.

The reference simulation reports approximately 300-V DC-bus regulation with approximately 10-ms recovery following a step input.

The battery voltage and current behaviour demonstrates successful transition between charging and discharging conditions.

Therefore, the MATLAB/Simulink implementation establishes the complete simulation framework required to evaluate the bidirectional DC/DC converter and its energy-management strategy.

6.27 Final Root Map for Chapter 6

The complete Chapter 6 development can be summarized through the following mathematical flow:

$$\boxed{\text{MATLAB/Simulink Implementation}} \quad (6.93)$$

$$\downarrow \quad (6.94)$$

$$\boxed{\text{Renewable Generation} + \text{DC Bus} + \text{Load} + \text{Battery}} \quad (6.95)$$

$$\downarrow \quad (6.96)$$

$$\boxed{\Delta P = P_G - P_L} \quad (6.97)$$

$$\downarrow \quad (6.98)$$

$$\boxed{\text{SOC Estimation}} \quad (6.99)$$

$$\downarrow \quad (6.100)$$

$$\boxed{\text{Energy Management}} \quad (6.101)$$

$$\downarrow \quad (6.102)$$

$$\begin{cases} P_G > P_L \Rightarrow \text{BUCK} \Rightarrow VT_2 \Rightarrow \text{Battery Charging} \\ P_G < P_L \Rightarrow \text{BOOST} \Rightarrow VT_1 \Rightarrow \text{Battery Discharging} \end{cases}$$

(6.103)

↓

(6.104)

PWM Generation → Switching Protection → Bidirectional DC/DC Converter

(6.105)

↓

(6.106)

Battery → $[V_B, I_B, SOC]$

(6.107)

↓

(6.108)

DC-Bus Regulation → $V_{DC} \approx 300 \text{ V}$

(6.109)

↓

(6.110)

Performance Evaluation

(6.111)

↓

(6.112)

Reference vs. Present Simulation

(6.113)

Chapter 7

Simulation and Results

7.1 Introduction

This chapter presents the MATLAB/Simulink implementation and simulation results of the bidirectional DC/DC converter based energy-storage system developed in the previous chapters.

The simulation is used to verify the operating principle of the bidirectional converter, the buck and boost control structures, the battery charging and discharging operation, and the resulting battery voltage and current waveforms.

The simulation model contains the complete energy-storage system together with the control circuits required for buck and boost operation. The original MATLAB/Simulink models supplied with the reference material are retained in this chapter without modification.

The simulation results are particularly important because they provide a direct connection between the mathematical equations developed in the previous chapters and the actual time-domain behaviour of the converter.

The overall relationship can therefore be represented as

$$\boxed{\text{Mathematical Model} \longrightarrow \text{Converter Design} \longrightarrow \text{Simulink Implementation} \longrightarrow \text{Simulation Results}} \quad (7.1)$$

The principal results considered in this chapter are:

1. Buck-mode control implementation.
2. Boost-mode control implementation.
3. Complete energy-storage system simulation.
4. Battery voltage waveform.
5. Battery current waveform.
6. Comparison of battery voltage and current during charging and discharging.
7. Verification of the calculated converter inductances.

7.2 Simulation Model of the Energy Storage System

The complete simulation model of the energy-storage system is shown in Fig. 7.1.

The original simulation model contains the major subsystems associated with the bidirectional converter, battery, buck and boost control sections, switching logic, SOC-related control and the corresponding measurement blocks.

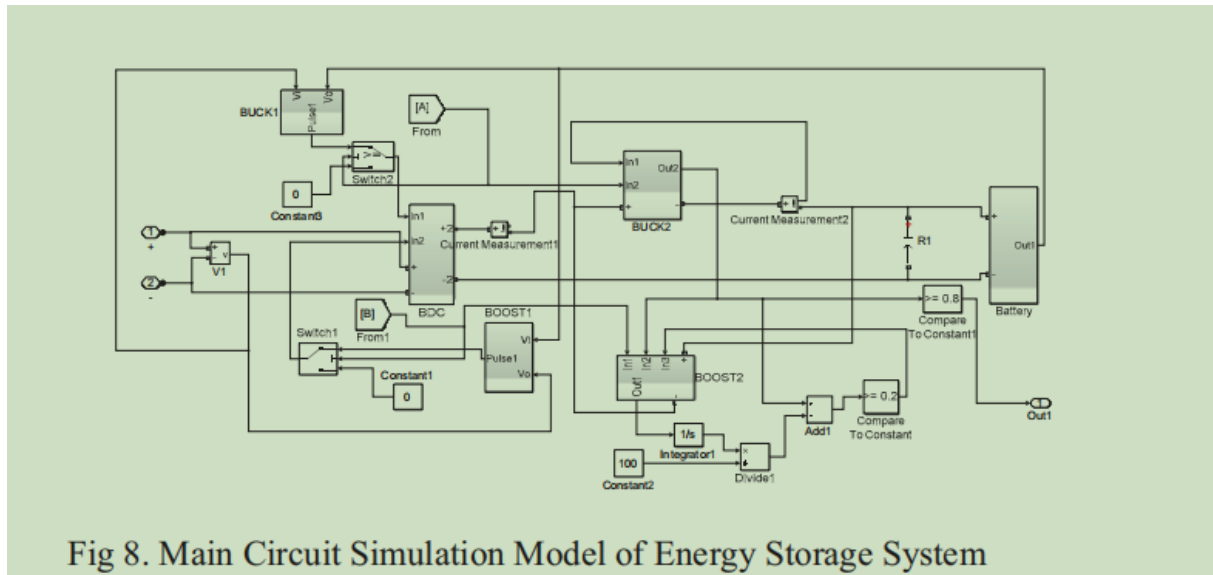


Figure 7.1: Original MATLAB/Simulink main circuit simulation model of the energy-storage system.

The complete model provides the simulation environment in which the mathematical relationships developed in the previous chapters are implemented.

In particular, the model combines the following functional blocks:

- DC-bus interface,
- battery subsystem,
- buck converter,
- boost converter,
- buck controller,
- boost controller,
- switching logic,
- current and voltage measurement,
- SOC-related logic.

Therefore, Fig. 7.1 represents the implementation-level realization of the theoretical converter architecture developed previously.

7.3 Buck Control Simulation

The original buck-control subsystem is shown in Fig. 7.2.

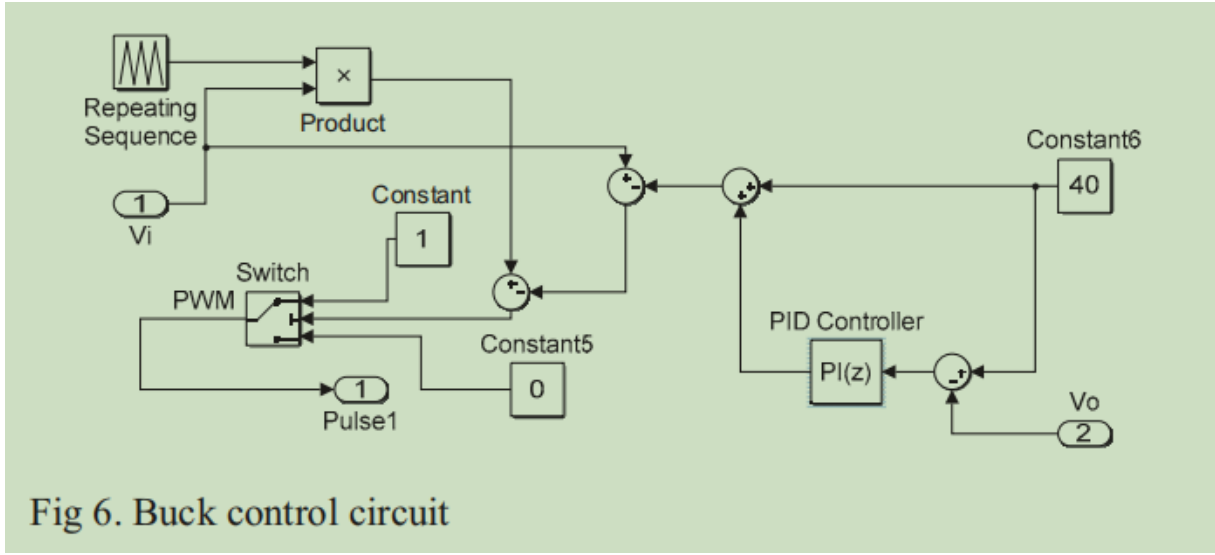


Fig 6. Buck control circuit

Figure 7.2: Original MATLAB/Simulink buck control circuit.

The buck controller regulates the converter during the charging condition. For buck operation, the ideal duty-ratio relationship is

$$D = \frac{U_o}{U_i} \quad (7.2)$$

where U_i is the input voltage and U_o is the required output voltage.

The control circuit uses the voltage measurement to determine the deviation from the desired operating condition. The controller then generates the control signal used for PWM switching.

The corresponding control sequence is

$$U_o^* \rightarrow \text{Error} \rightarrow \text{PID Controller} \rightarrow D \rightarrow \text{PWM} \rightarrow VT_2 \quad (7.3)$$

Therefore, the buck-control simulation shown in Fig. 7.2 provides the practical implementation of the theoretical buck duty-ratio relationship.

7.4 Boost Control Simulation

The original boost-control circuit is shown in Fig. 7.3.

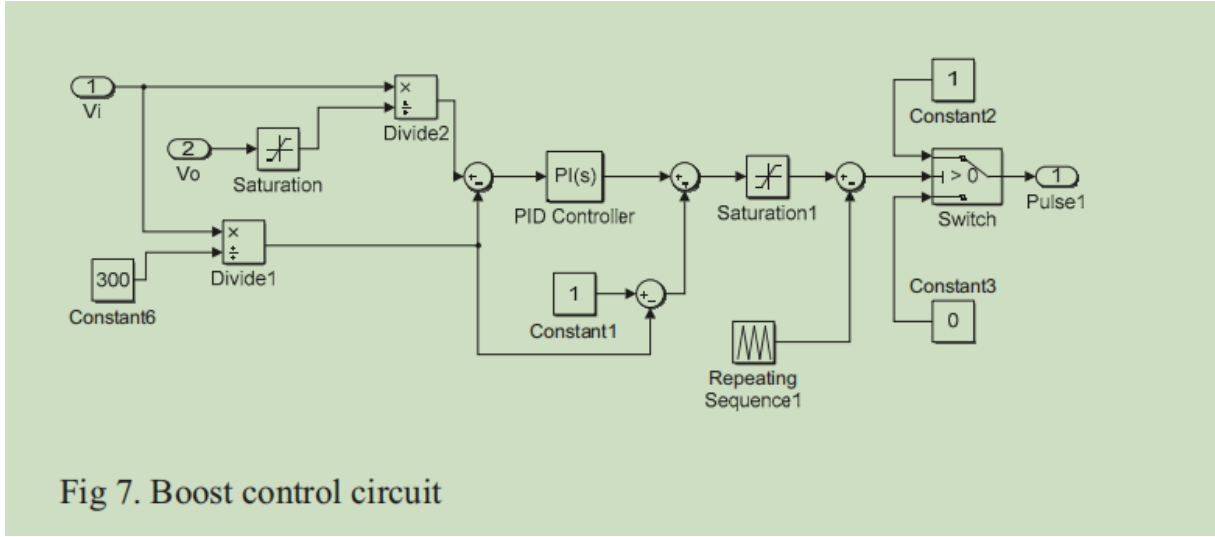


Figure 7.3: Original MATLAB/Simulink boost control circuit.

During battery discharging, the converter operates in boost mode. The ideal boost voltage relationship is

$$D = 1 - \frac{U_i}{U_o} \quad (5.1)$$

or equivalently,

$$U_o = \frac{U_i}{1 - D}. \quad (5.2)$$

The boost controller therefore calculates the required duty ratio according to the input and output voltage relationship and applies the required correction through the PID controller.

The practical implementation follows the sequence

$$U_i, U_o \rightarrow \text{Duty-Ratio Calculation} \rightarrow \text{PID Controller} \rightarrow D \rightarrow PWM \rightarrow VT_1 \quad (7.4)$$

The original Simulink implementation shown in Fig. 7.3 therefore provides the simulation realization of the boost-mode mathematical model.

7.5 Relation Between the Boost Model and Simulation

The mathematical model developed for the boost operating condition provides the basis for understanding the simulated converter response.

When VT_2 is OFF and VT_1 is ON, the converter operates in step-up mode. The corresponding small-signal model is described by the equations developed in the previous chapter.

The state equations can be written in the form

$$L \frac{d\hat{i}_L}{dt} = (D - 1)\hat{u}_o + \hat{d}U_o + U_i \quad (5.3)$$

and

$$C \frac{d\hat{u}_o}{dt} = (1 - D)\hat{i}_L - \hat{d}I_L - \frac{\hat{u}_o}{R}. \quad (5.4)$$

These equations show that the inductor current and output voltage are directly influenced by the duty-ratio perturbation $\hat{d}$.

Consequently,

$$\boxed{\hat{d} \rightarrow \hat{i}_L \rightarrow \hat{u}_o} \quad (7.5)$$

represents the fundamental control-to-output relationship of the boost converter.

The simulated boost controller in Fig. 7.3 implements this relationship through measurement, error processing, PID control and PWM generation.

7.6 Buck Inductor Design and Simulation

The buck-mode inductance is obtained from the design relationship

$$\boxed{L = \frac{U_o}{I_{pk}} \frac{1}{F_{sw}} \left(1 - \frac{U_o}{U_{i,\max}} \right)} \quad (5.5)$$

where

- U_o is the output voltage,
- I_{pk} is the peak inductor current,
- F_{sw} is the switching frequency,
- $U_{i,\max}$ is the maximum input voltage.

The calculated buck inductance used in the reference design is

$$\boxed{L_{\text{buck}} = 350 \mu\text{H}} \quad (5.6)$$

This value determines the energy-storage capability of the inductor during buck operation. The physical meaning of this relationship is important:

$$\boxed{L \rightarrow \text{Inductor Energy Storage} \rightarrow \text{Current Ripple} \rightarrow \text{Converter Response}} \quad (7.6)$$

Therefore, the 350 μH value is not simply a circuit parameter; it directly affects the current dynamics implemented in the simulation.

7.7 Boost Inductor Design and Simulation

For boost operation, the inductance is obtained from the design relationship

$$\boxed{L = \frac{U_i^2 D U_o \eta}{2f P_{out} (U_o - U_i \eta)}} \quad (5.7)$$

where

- U_i is the input voltage,
- U_o is the output voltage,
- D is the duty ratio,
- η is the converter efficiency,
- f is the switching frequency,
- P_{out} is the output power.

The resulting boost inductance is

$$L_{\text{boost}} = 104 \times 10^{-6} \text{ H} \quad (5.8)$$

or

$$L_{\text{boost}} = 104 \text{ } \mu\text{H}. \quad (5.9)$$

Thus, Eq. (5.7) directly determines the inductance implemented in the boost converter simulation. The relationship can be summarized as

$$\text{Boost Design Equation} \rightarrow 104 \text{ } \mu\text{H} \rightarrow \text{Simulink Inductor} \rightarrow \text{Boost Current Dynamics} \quad (7.7)$$

7.8 Regulator Circuit

The original regulator circuit supplied with the simulation model is shown in Fig. 7.4.

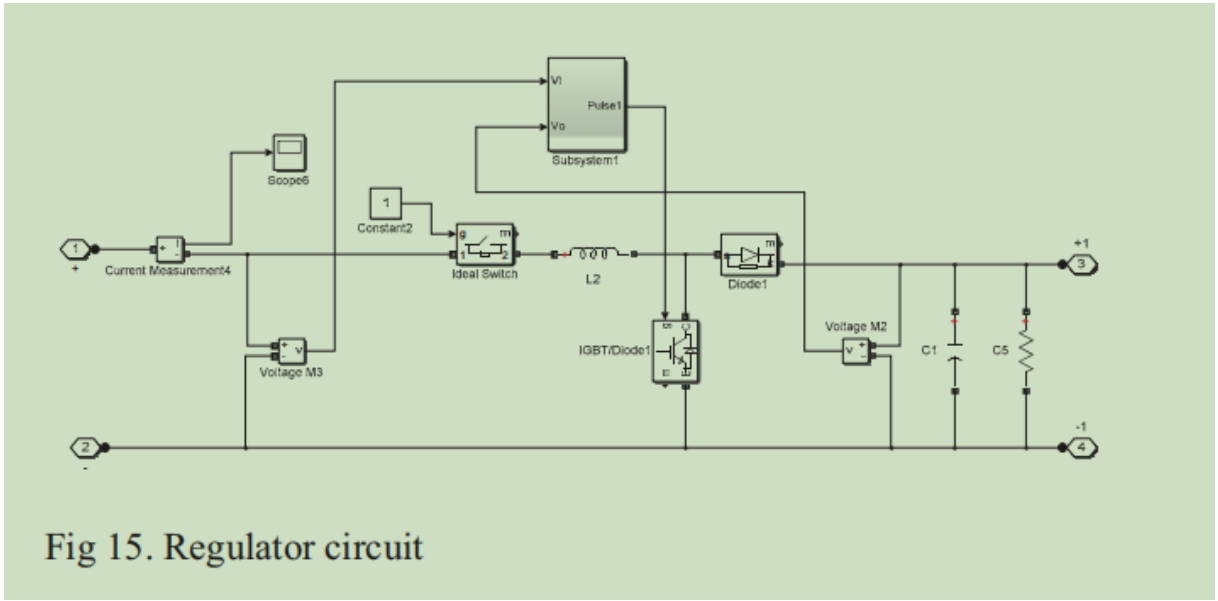


Figure 7.4: Original regulator circuit used in the simulation model.

The regulator circuit contains the switching device, inductive element, diode, capacitors and measurement elements associated with the power-conversion stage.

This subsystem connects the control signal generated by the controller to the actual electrical power stage.

Therefore, the overall simulation relationship is

$$\text{Controller} \rightarrow \text{PWM} \rightarrow \text{Switch} \rightarrow \text{Inductor} \rightarrow \text{Output} \quad (7.8)$$

7.9 Battery Voltage Simulation Result

The simulated battery voltage waveform is shown in Fig. 7.5.

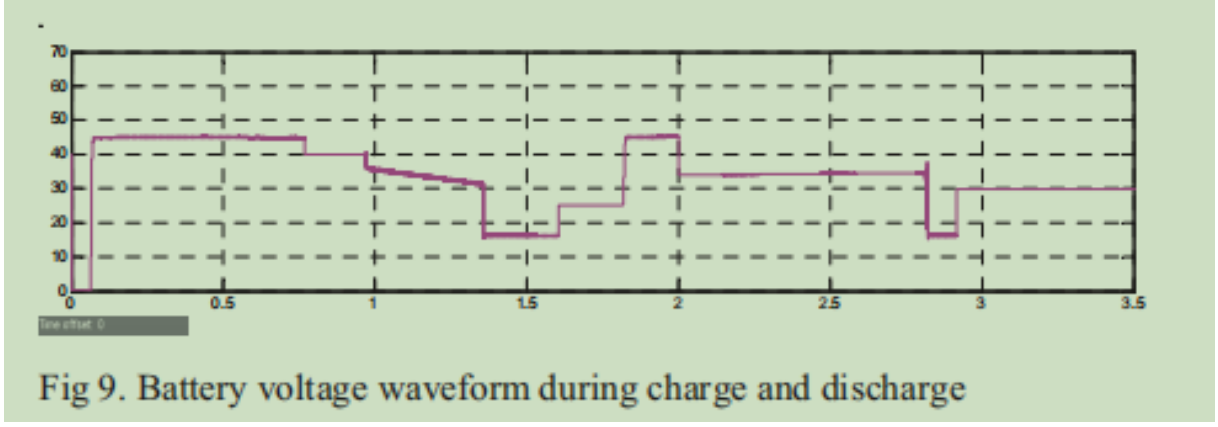


Figure 7.5: Original simulated battery voltage waveform during charging and discharging.

The waveform demonstrates that the battery voltage changes according to the operating condition of the energy-storage system.

The battery voltage exhibits different levels during the charging and discharging intervals. These changes are associated with the power-flow direction controlled by the bidirectional converter.

The relationship between the converter operation and battery voltage can therefore be expressed as

$$\boxed{\text{Operating Mode} \rightarrow D \rightarrow \text{Converter Voltage Conversion} \rightarrow V_B} \quad (7.9)$$

where V_B represents the battery voltage.

The voltage waveform therefore provides a time-domain indication that the converter is changing its operating condition according to the energy-storage requirement.

7.10 Battery Current Simulation Result

The original battery-current waveform is shown in Fig. 7.6.

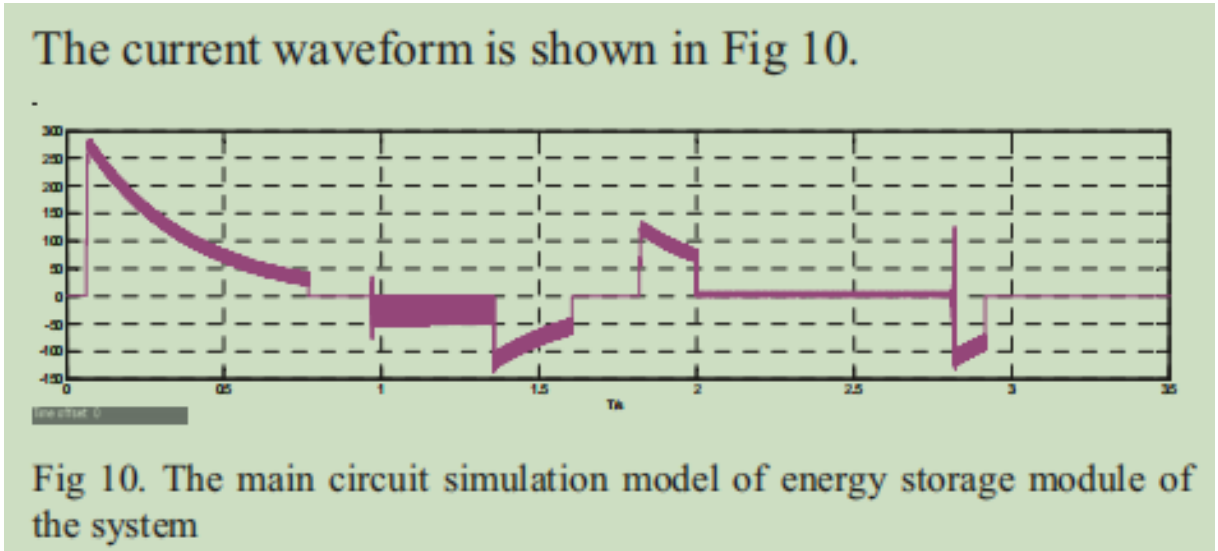


Figure 7.6: Original simulated battery current waveform of the energy-storage system.

The battery current is particularly useful for identifying the direction of energy transfer.

The sign and magnitude of the battery current change according to whether the battery is being charged or discharged.

The general relationship can be represented as

$$\boxed{P_B = V_B I_B} \quad (5.10)$$

where

- P_B is battery power,
- V_B is battery voltage,
- I_B is battery current.

Therefore, changes in the simulated battery current correspond directly to changes in battery power.

The current waveform also reflects the dynamic behaviour of the converter inductor because the battery current is linked to the energy-transfer path containing L .

7.11 Comparison of Battery Voltage and Current

The combined battery-voltage and battery-current simulation result is shown in Fig. 7.7.

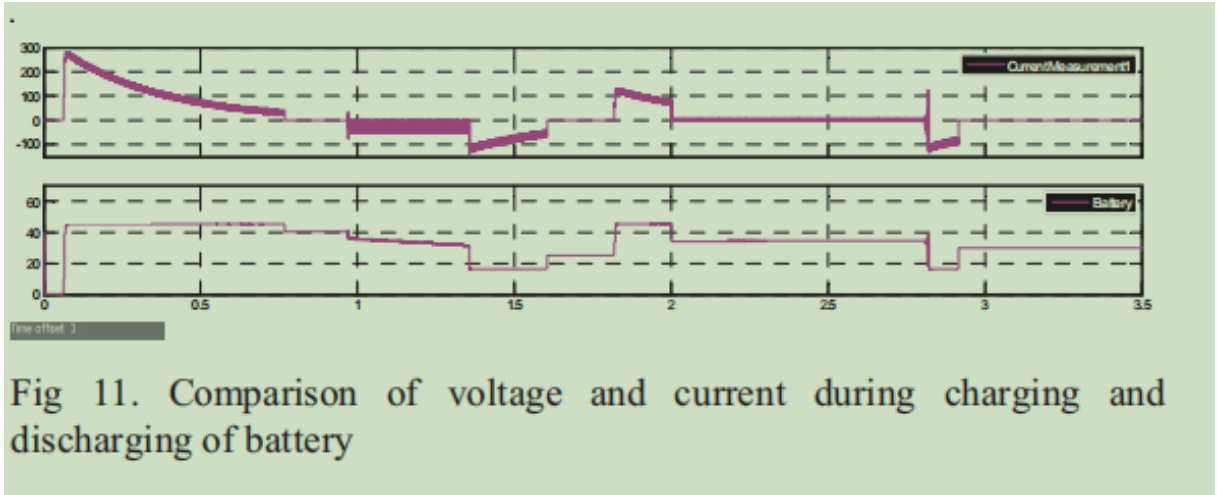


Figure 7.7: Original comparison of battery voltage and current during charging and discharging.

The comparison is important because voltage and current together determine the battery power:

$$P_B = V_B I_B. \quad (5.11)$$

Consequently, the two waveforms should not be interpreted independently.

The sequence of operation is

$$P_G - P_L \rightarrow \text{Charge/Discharge Decision} \rightarrow \text{Converter Mode} \rightarrow D \rightarrow V_B, I_B \quad (7.10)$$

Thus, the simulation results provide a direct visualization of the power-flow decision discussed in the previous chapters.

7.12 Main Circuit Current Waveform

The complete current waveform obtained from the main circuit simulation is shown in Fig. 7.8.

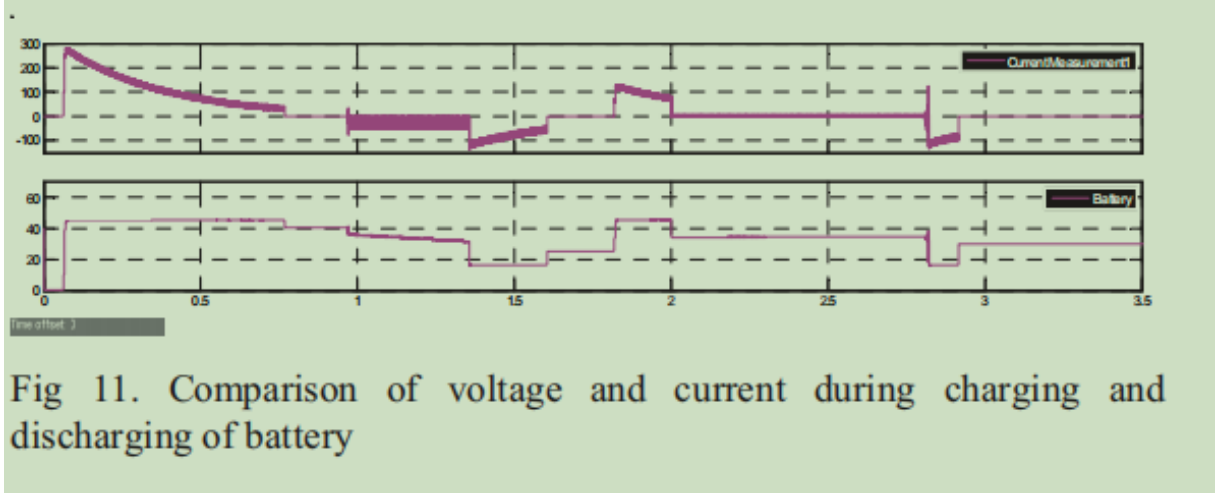


Fig 11. Comparison of voltage and current during charging and discharging of battery

Figure 7.8: Original current waveform obtained from the main circuit simulation.

The waveform contains different operating intervals corresponding to changes in the converter operating condition.

The inductor and battery current behaviour is directly related to the converter state equations.

For example, in the boost operating condition,

$$L \frac{d\hat{i}_L}{dt} = (D - 1)\hat{u}_o + \hat{d}U_o + U_i. \quad (7.11)$$

Therefore, changes in duty ratio influence the rate of change of inductor current.

The simulation waveform provides the corresponding time-domain response of this relationship.

7.13 Identification of Charging and Discharging Intervals

The simulated waveforms can be interpreted according to the power balance of the hybrid energy-storage system.

The charging condition is

$$\boxed{P_G > P_L} \quad (7.12)$$

where excess renewable power is available.

The corresponding energy flow is

$$\boxed{DC \text{ Bus} \rightarrow Converter \rightarrow Battery.} \quad (7.13)$$

The discharging condition is

$$\boxed{P_G < P_L} \quad (7.14)$$

and the energy flow becomes

$$\boxed{Battery \rightarrow Converter \rightarrow DC \text{ Bus}.} \quad (7.15)$$

Therefore, the simulated voltage and current waveforms should be interpreted together with the energy-management condition.

The complete relationship is

$$\boxed{\begin{array}{c} P_G > P_L \\ \downarrow \\ \text{Charging} \\ \downarrow \\ \text{Buck} \\ \downarrow \\ VT_1 = \text{OFF}, VT_2 = \text{ON} \end{array}} \quad (7.16)$$

and

$$\begin{array}{c}
 P_G < P_L \\
 \downarrow \\
 \text{Discharging} \\
 \downarrow \\
 \text{Boost} \\
 \downarrow \\
 VT_1 = \text{ON}, VT_2 = \text{OFF}
 \end{array}
 \quad (7.17)$$

7.14 Relation Between Mathematical Equations and Simulation Results

The simulation results are the practical realization of the mathematical equations derived in the previous chapters.

The principal relationships are summarized below.

7.14.1 Buck Duty-Ratio Equation

The ideal buck relationship is

$$D = \frac{U_o}{U_i}. \quad (5.12)$$

This equation determines the nominal duty ratio required to obtain the desired output voltage. In the simulation, this relationship is implemented through the buck controller shown in Fig. 7.2. Therefore,

$$D_{\text{buck}} \rightarrow PWM \rightarrow VT_2 \rightarrow V_B \quad (7.18)$$

during charging.

7.14.2 Boost Duty-Ratio Equation

For boost operation,

$$D = 1 - \frac{U_i}{U_o}. \quad (5.13)$$

The boost controller implements this relationship and modifies the duty ratio according to the voltage error.

Consequently,

$$D_{\text{boost}} \rightarrow PWM \rightarrow VT_1 \rightarrow V_{DC} \quad (7.19)$$

during battery discharging.

7.14.3 Small-Signal Model

The small-signal model relates the duty-ratio perturbation to inductor current and output-voltage perturbations.

The relationship

$$\hat{d} \rightarrow \hat{i}_L \rightarrow \hat{u}_o \quad (7.20)$$

explains why the simulated output voltage changes when the control signal changes.

Therefore, the voltage waveforms in Fig. 7.5 are the time-domain result of the dynamic relationships represented by the converter equations.

7.14.4 Inductor Design Equation

The buck inductor is obtained from

$$L_{\text{buck}} = 350 \mu\text{H}. \quad (7.21)$$

The boost inductor is obtained from

$$L_{\text{boost}} = 104 \mu\text{H}. \quad (7.22)$$

These values determine the energy-storage characteristics of the converter.
Thus,

$$L \rightarrow \frac{di_L}{dt} \rightarrow i_L(t) \rightarrow P_B(t). \quad (7.23)$$

This provides the direct connection between the inductance-design equations and the current waveform obtained from simulation.

7.15 Simulation Validation

The simulation can be used to validate the theoretical development at several levels.

7.15.1 Topology Validation

The main simulation model confirms that the theoretical converter topology has been implemented in MATLAB/Simulink.

This is demonstrated by Fig. 7.1.

7.15.2 Control Validation

The original buck and boost control circuits shown in Figs. 7.2 and 7.3 demonstrate the implementation of the two operating modes.

7.15.3 Parameter Validation

The inductance values calculated from the design equations are implemented in the corresponding converter models:

$$L_{\text{buck}} = 350 \mu\text{H} \quad (7.24)$$

and

$$L_{\text{boost}} = 104 \mu\text{H}. \quad (7.25)$$

7.15.4 Waveform Validation

The battery voltage and current waveforms shown in Figs. 7.5, 7.6, and 7.7 demonstrate the time-domain behaviour resulting from the implemented converter and control system.

Therefore, the simulation establishes the following chain:

$$\text{Equations} \rightarrow \text{Parameters} \rightarrow \text{Control} \rightarrow \text{Simulation} \rightarrow \text{Waveforms}. \quad (7.26)$$

7.16 Simulation Results Summary

The principal simulation observations are summarized in Table 7.1.

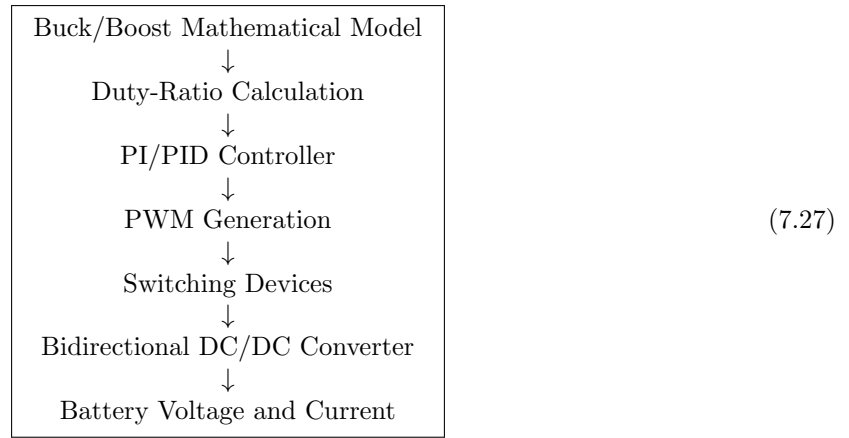
Table 7.1: Summary of simulation implementation and results.

Aspect	Theoretical Basis	Simulation Evidence
Buck operation	$D = U_o/U_i$	Original buck control circuit and charging operation
Boost operation	$D = 1 - U_i/U_o$	Original boost control circuit and discharging operation
Buck inductance	Buck inductance design equation	$L_{\text{buck}} = 350 \mu\text{H}$ implemented in the converter model
Boost inductance	Boost inductance design equation	$L_{\text{boost}} = 104 \mu\text{H}$ implemented in the converter model
Battery voltage	Converter voltage conversion	Original battery-voltage waveform
Battery current	Inductor/current dynamics and $P_B = V_B I_B$	Original battery-current waveform
Charging/discharging	$P_G - P_L$	Combined voltage/current waveform
PWM switching	Duty ratio and switching frequency	Original Simulink control implementation

7.17 Overall Result Interpretation

The simulation results demonstrate the complete connection between the mathematical model, converter design and practical simulation implementation.

The overall relationship can be summarized as



The simulation results therefore provide the final time-domain representation of the theoretical converter behaviour.

In particular, the inductance equations determine the converter energy-storage elements, the duty-ratio equations determine the nominal switching requirements, and the control circuits modify the switching command according to the measured electrical quantities.

The resulting battery voltage and current waveforms then reflect the response of the complete closed-loop system.

7.18 Chapter Conclusion

This chapter presented the MATLAB/Simulink simulation and results of the bidirectional DC/DC converter based energy-storage system.

The original simulation model was presented together with the original buck and boost control circuits. The calculated buck and boost inductances were related directly to their respective converter-design equations.

The simulated battery-voltage and battery-current waveforms were then analyzed in relation to the converter operating conditions.

The key relationships established in this chapter are

$$D_{\text{buck}} = \frac{U_o}{U_i} \quad (7.28)$$

and

$$D_{\text{boost}} = 1 - \frac{U_i}{U_o} \quad (7.29)$$

together with

$$L_{\text{buck}} = 350 \mu\text{H} \quad (7.30)$$

and

$$L_{\text{boost}} = 104 \mu\text{H}. \quad (7.31)$$

These theoretical relationships are implemented in the simulation through the buck and boost control circuits, PWM generation and switching stages.

The final battery voltage and current waveforms provide the time-domain evidence of the resulting converter operation.

Thus, the simulation chapter establishes the complete path from **mathematical derivation** to **converter design**, **control implementation**, and finally to **simulation results**.

Simulation and Results Root Map

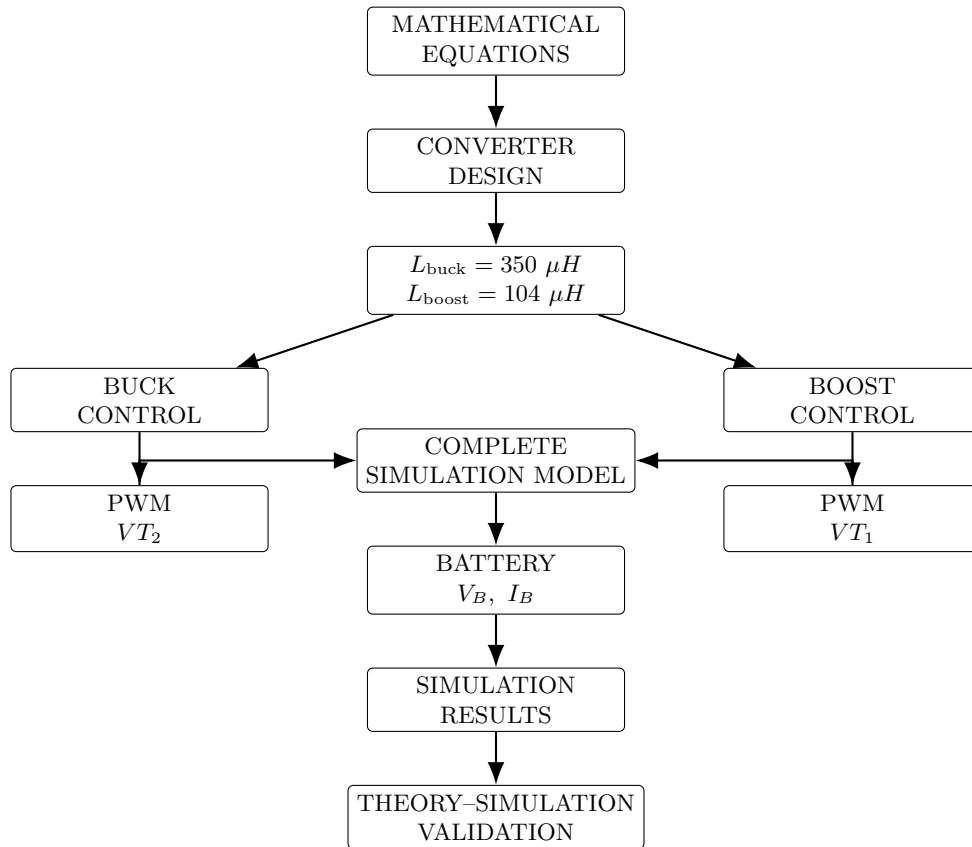


Figure 7.9: Root map connecting mathematical equations, converter design, control implementation and simulation results.